ROOT CAUSE UNLOCKED

ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your *GLP-1*, Ultimate Guide

Keep the Weight Loss. Keep Your Strength, Your Skin, and Your Long-Term Health, Too.

A guide for adults losing weight on semaglutide or tirzepatide (Ozempic, Wegovy, Mounjaro, Zepbound) who want to understand, and address, the cellular side of rapid fat loss that almost nobody explains at the pharmacy counter.

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BEFORE YOU START

> ***Disclaimer:*** *This book is for educational purposes only and does not constitute medical advice. It does not diagnose any condition, and reading it does not make you my patient. Always consult a qualified healthcare provider before starting any new treatment, supplement, or dietary change, and do not start, stop, or change any prescribed medication without speaking to the prescriber. Keep taking your GLP-1 medication exactly as prescribed. If you have had persistent vomiting or months of very low intake and then develop confusion, unsteadiness, or vision changes, treat that as an emergency and read Key 5 now.*

A note before you start. This guide is educational information. It explains the biology behind a documented consequence of rapid GLP-1 driven weight loss and the general categories of intervention used to address it. It is not individualized medical advice, and it does not replace a relationship with your prescribing physician or a clinical evaluation with my team. Keep taking your GLP-1 medication exactly as prescribed. Talk to your doctor or a qualified practitioner before starting any new supplement, especially if you take blood thinners, thyroid medication, diuretics, or other prescriptions.

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INTRODUCTION

What You Are About to Learn

If you are losing weight on a GLP-1 medication, it is working. The appetite reduction is real, the number on the scale is moving, and the metabolic benefits are genuine and well documented. That is not in question.

Here is what usually is not covered in a fifteen-minute prescribing appointment: the speed of that weight loss creates a cellular consequence that has nothing to do with your willpower, your diet, or your age, and everything to do with how fast your body is mobilizing fat. It is a mechanism, not a mystery, and it is documented in peer-reviewed research. Once you understand it, you can do something about it while you keep taking your medication exactly as prescribed.

This guide exists so you can keep the result you came for, the weight loss, without quietly trading away the muscle, the skin elasticity, the energy, and the long-term cellular health that most people do not realize are on the table. The goal is not to talk you out of your medication. The goal is that you go into this with your eyes open, protecting your strength and your energy on the way down rather than discovering a year from now that you lost more than fat. I am not going to tell you how much better you will feel or by when. What I can tell you is what is worth protecting and how.

Below are eight keys. Each one explains a specific consequence of rapid GLP-1 driven fat loss, the mechanism behind it in plain language, and what you can do about it at an educational level. The last section walks through the structured 90-day program I built to address all of this directly, if you decide you want a guided, lab-directed approach rather than piecing it together yourself.

KEY 1

The Weight Is Coming Off. So Is Something Else.

The problem nobody mentions. Your GLP-1 medication is doing exactly what it was designed to do. But when fat tissue breaks down as fast as it does on these medications, it releases fatty acids into your bloodstream faster than your mitochondria can process them. That overload has a name and a documented downstream effect that your prescriber's monitoring panel is not built to catch.

The mechanism, in plain terms. GLP-1 receptors do not only sit on the pancreas and the appetite centers in your brain, the places your medication is meant to act. They are also present on adipose-derived stem cells and the fibroblasts in your skin.[1] When your medication stimulates those receptors, it reduces the protective cytokines those stem cells normally produce to shield nearby fibroblasts from oxidative damage.[2] At the same time, the fatty acid flood from rapid fat mobilization generates a surge of reactive oxygen species, unstable molecules that damage DNA, break down collagen, and overwhelm your natural antioxidant defenses. This combination is documented in a 2025 peer-reviewed paper on GLP-1 receptor agonists and skin aging,[1] and separately in 2026 research on the metabolic and redox effects of GLP-1 therapy.[3]

Cells that absorb too much of this oxidative damage do not simply die and get cleared away. They enter a state called senescence: they stop functioning, stop dividing, but refuse to die. Researchers call them zombie cells. Instead of going quietly, senescent cells release a continuous stream of inflammatory signaling molecules known as SASP, the Senescence-Associated Secretory Phenotype.[4]

Where the certainty stops, stated plainly. The senescence and SASP biology above is well established in the laboratory. The step from that biology to the specific symptoms in the next two keys, in a specific person on a GLP-1, is a proposed mechanism rather than a proven chain. Nobody has run the trial that would settle it. I use this model clinically because it fits what I see and because everything it points to (protein, training, nutrient repletion, clearance) is low risk and defensible on its own terms. You should know which of those two things you are buying.

What to do about it, at an educational level. This is not a reason to stop your medication but a reason to add a second, complementary layer of support that addresses the cellular side effect directly: clearing the senescent cells that have already formed, rebuilding the antioxidant reserves that were depleted, and giving your body the structural building blocks (protein, collagen, lymphatic support) it needs to keep pace with how fast you are losing weight. The rest of this guide walks through each of those pieces.

KEY 2

Up to Half the Weight You Lose Might Be Muscle, Not Fat

The half of weight loss nobody warns you about. Most people assume weight loss on a GLP-1 medication is fat loss. It is not entirely.[5] A landmark study on semaglutide published in Cell Metabolism found that in the STEP-1 trial, lean mass accounted for roughly 45 percent of total weight lost, nearly double the rate that would be expected from caloric restriction alone. That is muscle, and once it is gone, it does not come back just because you stopped losing weight.

Why muscle goes, in plain language. The muscle loss associated with GLP-1 therapy is not caused by one single thing. Research reviewing the pattern across GLP-1 patients points to several factors working together: reduced protein intake because appetite suppression makes it easy to under-eat protein specifically, a negative energy balance that pushes your body toward breaking down tissue for fuel, hormonal shifts, impaired signaling through the mTOR pathway that normally tells your body to build and maintain muscle, and increased protein breakdown.[6] For patients on tirzepatide specifically, there is additional evidence that the GIP component of the medication can push muscle precursor cells toward becoming fat cells instead of muscle, compounding the effect.

The consequence is not cosmetic. Reduced grip strength, functional decline, and accelerated age-related muscle loss (sarcopenia) beyond what caloric restriction alone would explain have all been documented in GLP-1 treated patients, including in a 24-month retrospective study specifically measuring muscle mass, grip strength, and gait speed.[7]

The two levers that matter, at an educational level. Two things counter this mechanism directly, and they are the two most important actions in this entire guide: protein and resistance training. Your target is 1.6 to 2.0 grams of protein per kilogram of body weight per day, distributed across three to four meals rather than back-loaded into one. Because appetite suppression makes this hard, prioritize protein first at every meal, before anything else. Pair that with resistance training a minimum of three times per week, using compound movements (squats, rows, presses, hip hinges) worked to within two to three repetitions of failure, with a protein-rich meal or shake within two hours after each session.[8] This is the single most direct lever you have against the lean mass consequence of rapid GLP-1 driven weight loss, and it also independently activates a cellular cleanup process called autophagy, which works alongside the senolytic support described in Key 4.

A third, smaller lever: omega-3 fats. After protein and resistance training, one more input has real human trial evidence, at a much more modest level. A 2023 meta-analysis of randomized trials found that omega-3 supplementation produced a very small but statistically significant benefit to muscle strength, with no meaningful effect on muscle mass or function overall.[25] The most encouraging single trial was run in exactly the group with the most to lose: healthy older adults, aged 60 to 85, who took about 3.4 grams of combined EPA and DHA daily for six months and gained 3.6 percent thigh muscle volume, 2.3 kilograms of grip strength, and 4 percent on one-rep-max strength compared with the control group.[26] Keep the size of this in perspective. Omega-3 is a garnish, not the meal, and nothing about it replaces the protein target or the training. But if you are an older adult on a GLP-1, or you already take fish oil for other reasons, the evidence suggests it is quietly working in your favor. The doses studied were in the low single-digit grams of combined EPA and DHA per day, which is considerably more than a standard one-capsule fish oil provides, so check your label. And as with everything in this guide, clear any new supplement with your prescriber first, especially if you take a blood thinner.

KEY 3

Why Your Skin, Joints, and Energy Feel Off, and Your Prescriber Cannot Explain It

The symptoms your monitoring panel will not catch. Accelerated skin aging or looseness disproportionate to the weight lost, sometimes called "Ozempic face." New joint pain or diffuse achiness that started after you began your medication. Fatigue that does not track with your improving blood sugar or cholesterol numbers. Brain fog, word-finding trouble, or memory changes. Increased hair shedding. These symptoms typically show up somewhere between three and twelve months into GLP-1 therapy, and they are frequently written off as aging, inadequate nutrition, or "just how weight loss feels." Standard GLP-1 monitoring panels, the ones checking blood sugar and lipids, are not built to catch what is actually driving them.

One proposed source behind all of it. The proposed common source for this whole cluster is the senescent, or "zombie," cells described in Key 1 and the SASP inflammatory signal they release continuously. Laboratory work indicates that SASP cytokines degrade the extracellular matrix that supports your skin and joints, which is the proposed mechanism behind both the accelerated skin aging and the new joint discomfort. SASP also induces an enzyme called CD38 in neighboring healthy cells, which depletes their NAD+, a molecule your cells depend on for energy production.[9] That is a plausible contributor to the fatigue and brain fog that do not improve even as your metabolic numbers do, and I want to be clear that plausible is not the same as demonstrated in patients like you. The hair changes trace back to the same cellular stress and nutritional depletion that comes with rapid weight loss on a reduced-appetite diet.

A different way to read these symptoms, at an educational level. The important reframe here is that this symptom cluster is not an inevitable cost of GLP-1 therapy, and it is not "just aging." There is a specific and plausible process behind it, and every lever it points to is addressable. It responds to the same interventions covered in the rest of this guide: Key 4 covers clearing the senescent cells generating the SASP signal, Key 5 covers rebuilding the antioxidant and NAD+ reserves that were depleted, and Key 6 covers supporting the lymphatic clearance pathways that remove the inflammatory debris those cells leave behind. If any of these symptoms are severe or worsening rather than mild and stable, that is worth a direct conversation with your prescribing physician, not just a supplement.

KEY 4

Clearing the Zombie Cells

A burden that only grows. Senescent cells do not resolve on their own. Once a cell enters senescence, it stays there, continuing to release inflammatory SASP signals indefinitely, and it can even nudge nearby healthy cells toward senescence too. The longer they accumulate, the larger the inflammatory burden becomes. This is why the symptoms in Key 3 tend to build gradually rather than appear and resolve on their own.

How senolytics work, simply put. A category of natural compounds called senolytics has been studied specifically for their ability to selectively clear senescent cells while leaving healthy cells alone. In a screening of candidate compounds conducted at the Mayo Clinic, fisetin, a flavonoid found in strawberries and other fruits, was identified as the most potent natural senolytic tested.[10] Quercetin and curcumin are studied alongside it for complementary antioxidant and anti-inflammatory effects on the same pathway.

One detail matters clinically: senolytics do not work like a typical daily supplement. Research on senolytic mechanisms describes a “hit-and-run” effect, meaning the compounds do their work in a short window and then can be stopped, rather than needing continuous daily exposure to be effective.[11] This is why senolytic protocols are typically built around short pulsed cycles rather than everyday dosing.

What you are buying here, stated plainly. The senolytic evidence is laboratory work, animal work, and a small number of early human studies. There are no human outcome trials showing that fisetin, quercetin, or curcumin improve how a GLP-1 patient feels or functions, and I could not find one. That is what happens to compounds nobody can patent: the trials never get funded, and the marketing fills the gap. So what this section describes is reasoned clinical practice held to an endpoint, not trial-proven certainty. Two consequences worth acting on. Put your money into the protein and the resistance training from Key 2 first, because those are the levers with real human evidence behind them. And hold anything you add here to a defined endpoint rather than running it indefinitely.

Using this wisely, at an educational level. If you are exploring senolytic support, understand that timing matters as much as the compound itself. Continuous daily high-dose use is not how these compounds were studied, and pulsed cycling is what the mechanism research points toward. This is also why senolytic support is best paired with the antioxidant and lymphatic support covered in the next two keys: the proposed model has clearing cells release their stored inflammatory contents one final time, and your body needs the capacity to process and clear that debris.

KEY 5

Refilling the Tank, and Why Guessing at Supplements Wastes Money

A double depletion that gets half the attention. Two things are draining at once, and only one of them gets talked about. The reactive oxygen species overload described in Key 1 draws down your antioxidant reserves, specifically glutathione and CoQ10, faster than diet alone typically replaces them. But there is a second, simpler drain that matters more and gets mentioned less: you are eating dramatically less food. In the trial that worked out how semaglutide actually produces weight loss, total energy intake across a day fell by 24 percent.[14] Your requirement for vitamins and minerals did not fall by 24 percent. Reviews of GLP-1 therapy now describe a recognizable pattern of micronutrient shortfalls that follow from sustained low intake, and they are not evenly distributed. Not every patient is short in the same places, or to the same degree.[13]

What is actually running low. Glutathione is your body's primary intracellular antioxidant, and GGT (an enzyme measurable on routine bloodwork) rises when glutathione turnover is under active oxidative stress. GLP-1 therapy also reduces circulating cholesterol and alters lipid flux in ways that are biologically linked to reduced CoQ10 and glutathione peroxidase levels, the mitochondrial cofactors your cells need for energy production. And as covered in Key 3, the SASP signal from senescent cells actively depletes NAD+ in surrounding healthy cells through CD38 induction.[9]

Thiamine, vitamin B1, deserves its own paragraph, because it behaves differently from the others. Your body holds only a small reserve of it relative to what it uses daily, so thiamine status falls within weeks of poor intake rather than over months. That timeline lines up uncomfortably well with GLP-1 therapy, particularly if nausea or vomiting has been part of your experience. This is not hypothetical for people who look like you: in one obesity medicine clinic population, about a third of patients already tested thiamine deficient before any surgery or medication.[17] Low thiamine is also common in type 2 diabetes.[18] Bariatric surgery creates the same combination of sharply reduced intake plus vomiting, and precisely because of that, it has an established thiamine monitoring protocol built around it.[16] Reviews are now asking directly whether GLP-1 care should borrow that framework rather than reinvent it.[15]

Seek care now, not later: the one emergency in this guide. If you have had persistent vomiting or months of very low intake on a GLP-1 medication, and you then develop confusion or unusual mental fuzziness, unsteadiness on your feet, or vision changes such as double vision or trouble moving your eyes, that combination is a medical emergency and not a supplement problem. It describes Wernicke's encephalopathy, which is caused by thiamine deficiency. Go to an emergency department the same day and say the words "possible thiamine deficiency," because treatment is intravenous thiamine and it needs to happen before the damage sets.

Keep this in proportion. It is rare. A systematic review of published cases found six reported with semaglutide,[19] and an analysis of the FDA adverse event database identified fifteen cases across GLP-1 medications as a whole.[20] Set against many millions of prescriptions, that is a very small number, and it is not a reason to stop your medication. But two details from those reports are worth carrying with you. Nearly every patient had prolonged gastrointestinal symptoms or substantial rapid weight loss first, meaning there was a warning period. And among patients followed afterward, lasting neurological problems were common, which is what makes early recognition matter so much. Caught early it is treatable. Missed, it may not be reversible.

The lab-first approach, at an educational level. The most useful mindset here is one I apply clinically: the most expensive supplement is the one you do not actually need. Rather than assembling a long list of antioxidant products by guesswork, the more precise approach is to find out, through bloodwork, what is actually depleted in your body and direct support specifically there. Thiamine is measurable, which puts it firmly in the "test, do not guess" category alongside the rest. This is exactly what lab-directed protocols are built to do, and it is a core piece of the structured program described at the end of this guide. If you are not working with a practitioner yet, the general categories worth knowing about are a glutathione precursor (NAC) paired with direct glutathione support, a CoQ10 source for mitochondrial support, and thiamine if your intake has been low for a sustained stretch or if nausea and vomiting have been a recurring part of your experience.

A word about NAD+ boosters, since they are marketed heavily to you. The underlying biology is real: senescent cells do deplete NAD+ in the tissue around them, which is why NMN and similar products are pitched to GLP-1 patients for fatigue and low cellular energy.[9] The human evidence is thinner than the marketing suggests. The trials that exist are small and measure laboratory markers rather than outcomes people can feel or that predict health years later.[24] More relevant to you specifically, a 2024 analysis linked the terminal breakdown products of excess niacin to major adverse cardiac events across multiple patient cohorts, with laboratory work showing one of those metabolites directly triggers blood vessel inflammation.[23] That is an association plus a plausible mechanism. It is not proof that taking an NAD+ precursor harms anyone, and it should not be reported to you as though it were. But niacin, NMN, and related compounds are cleared through the same pathway, and if you are on a GLP-1 medication you are, almost by definition, someone whose cardiovascular risk is actively being managed. An unresolved question about blood vessel inflammation, set against a benefit that has not been demonstrated, is a poor trade in your situation. This guide does not recommend routine NAD+ precursor supplementation.

If you do supplement thiamine, which form. Benfotiamine is a fat-soluble thiamine derivative that raises thiamine levels in the body more effectively than standard thiamine hydrochloride.[21] Better absorption is not the same as better results. The longest trial run to date gave benfotiamine for twelve months to people with type 2 diabetes and symptomatic nerve pain. It reliably raised every thiamine marker measured and was well tolerated, but it produced no significant improvement in nerve structure, nerve function, or symptoms compared with placebo.[22] Read honestly, that means benfotiamine is a proven way to raise thiamine status and a poor bet as a treatment for established neuropathy. Raising status is the purpose that matters here, and benfotiamine is the form I use clinically: 150 mg twice daily for 4 to 6 weeks, run as a repletion course with a stop point rather than an open-ended habit, ideally with your level tested first so you know whether you needed it. No outcome trial has tested that course in people on a GLP-1 medication, and I would rather say so than imply one exists.

KEY 6

Clearing the Debris, Not Just Stopping the Damage

What happens to the debris. Clearing senescent cells and quenching oxidative stress is only half the job. Dying cells, cleared cellular debris, and inflammatory SASP byproducts have to actually leave your tissue. If that clearance pathway is sluggish, the debris lingers and can re-trigger the same inflammatory signaling you are trying to resolve.

Your body's drainage route, explained. Your lymphatic system is the clearance pathway responsible for removing this kind of cellular and inflammatory debris at the tissue level. In a structured cellular restoration protocol, lymphatic support is sequenced across the full course of treatment in phases, one remedy at a time with no overlap, specifically because it is the pathway responsible for clearing what the senolytic and antioxidant work leaves behind.

Keeping the pathway moving, at an educational level. This is the piece most people skip entirely because it is the least visible. Practical, low-cost support includes adequate hydration (a minimum of eight to ten glasses of filtered water daily), movement (even walking supports lymphatic flow, since unlike blood, lymph has no dedicated pump and relies on muscle contraction), and, if you are pursuing a structured protocol, dedicated lymphatic drainage support sequenced across the full course of your program rather than a single short course.

KEY 7

What to Actually Expect, Week by Week

Why the first weeks can mislead you. People starting a cellular restoration protocol alongside their GLP-1 medication often expect linear improvement from day one. It does not work that way, and not knowing this in advance is one of the most common reasons people quit a protocol that was, in fact, working exactly as expected.

The timeline, explained plainly. The model says that when senolytic support begins clearing senescent cells, those cells release their stored inflammatory contents one final time as they are eliminated. In the first one to two weeks, some people report temporary fatigue, mild joint achiness, or subtle brain fog, and the usual explanation offered is that clearance flush.

One caution about that framing, because it matters. If feeling better proves the protocol is working and feeling worse also proves it is working, the claim cannot be tested, and an untestable claim is a poor reason to keep paying. So I treat worsening as information, never as a milestone. Sometimes it is the flush and it passes. Sometimes it means the dose is too high, the lymphatic and hydration groundwork in Key 6 was skipped, or this is the wrong approach for you. Significant worsening is a reason to reassess with me, not a badge to wear.

The rest of the sequence is structured the same way: weeks three and four are where the early adjustment usually settles, weeks five through eight are where the antioxidant and mitochondrial work is most active, and weeks nine through twelve are the deepest phase of the same work. That is a description of what the protocol is doing in each window. It is not a forecast of how you will feel in each window, and I am not going to give you one.

Setting expectations, at an educational level. Take a baseline before you change anything: energy out of ten, sleep out of ten, and how you feel in the two days after a training session. Then judge everything against your own log rather than against anyone's promise, including mine. Symptoms that are severe, that are not improving after the first couple of weeks, or that appear or worsen later in the process (week four or beyond) are the signal to stop guessing and talk to a practitioner, since later-onset symptoms are handled differently than the expected early adjustment window. And if twelve weeks of your own log shows nothing moved, that is a real result. Stop paying for it.

KEY 8

This Runs Alongside Your Medication, Not Against It

The hesitation worth addressing. It is a reasonable question: if GLP-1 medications cause this cellular side effect, does addressing it mean working against your prescription? No. And understanding why is worth knowing, because it resolves a real, and reasonable, hesitation.

Two pathways, one medication. GLP-1 receptor activity itself has documented anti-aging potential at the receptor signaling level, including activation of AMPK and SIRT1 (pathways linked to cellular longevity), upregulation of NRF2 (a master regulator of your antioxidant defenses), promotion of autophagy, and suppression of NF-kB, a key inflammatory signaling pathway.[12] In other words, the same receptor your medication is designed to activate has real protective effects. The cellular aging consequence described throughout this guide is a separate, tissue-specific effect: GLP-1 receptor stimulation specifically on adipose-derived stem cells, compounded by the sheer speed of fat mobilization outpacing your mitochondria's capacity to process it. These two things, the receptor's genuine benefits and the fat-mobilization side effect, are happening at the same time, in different tissues, through different pathways. That is the paradox, and it is exactly why this guide exists: to address the specific downside without touching the benefit you are already getting.

The bottom line, at an educational level. Keep taking your GLP-1 medication exactly as your prescriber directs. Nothing in this guide is designed to replace, adjust, or interfere with that prescription. The interventions covered here (senolytic support, antioxidant rebuilding, lymphatic clearance, protein and resistance training) are additive. They run concurrently, targeting a documented side effect of the process your medication has already set in motion, so that the version of you on the other side of this weight loss is stronger and healthier, not just lighter.

The 90-Day GLP-1 Cellular Restoration Program

Everything in this guide describes the problem and the general categories of solution. The GLP-1 Cellular Restoration Program, which I built at Seay Wellness, is the structured, personalized version: a 90-day protocol that addresses all four mechanisms covered in this guide at once, senescent cell clearance, antioxidant restoration, physical clearance of cellular debris, and structural rebuild of lean mass, rather than leaving you to piece it together on your own.

The program is built on a core-and-enhanced structure. Every patient starts with a core protocol on day one, covering the foundational support that every GLP-1 patient benefits from regardless of individual lab results: omega-3s, magnesium, vitamin D, collagen peptides, lymphatic drainage support, and senolytic compounds on the correct pulsed schedule. Then, after a telehealth visit in week two where your actual lab results are reviewed, an enhanced layer is added, personalized to what your bloodwork shows you actually need. This is the "most expensive supplement is the one you do not need" principle from Key 5, put into practice instead of just explained.

Two telehealth consultations with me are built into every tier at no extra cost: an intake call that screens your full medication list for interactions before anything ships, and a week-two call where your labs are reviewed and your personalized protocol is finalized.

Foundation: the complete core-and-enhanced supplement protocol, shipped nationally to any US address, with your full patient protocol guide covering dosing, diet, and the resistance exercise protocol from Key 2. No in-office visits required.

Complete: everything in Foundation, plus in-office sessions at Seay Wellness in Garner, NC, combining photobiomodulation laser therapy, pneumatic lymphatic compression, infrared sauna, and neurological recovery support, along with a functional assessment (grip strength, posture, range of motion) at intake and again at the end, so you are comparing yourself against your own starting numbers rather than against a description from memory.

Elite: everything in Complete, plus a full lab panel at intake and again at the end, including biological age testing. A word on that one, because it is oversold everywhere else: biological age tests are an estimate built from a model, not a measurement of your cells, and the models disagree with each other. What it gives you is a consistent number you can run twice and compare to itself. That is genuinely useful as a before-and-after on your own record. It is not proof of anything, and I will not present it to you as proof.

If you have made it this far, you already understand more about what rapid GLP-1 driven weight loss is doing to your cells than most people ever will. The next step is finding out exactly where you stand.

Take the Next Step: Your Free Root Discovery Session

The Root Discovery Session is a complimentary consultation with my team to review your specific situation: how long you have been on your medication, what symptoms you are noticing, and which tier of the program fits where you are right now. There is no obligation, and if you decide to move forward, this conversation is simply the first step.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, start your free tracker at rootcauseunlocked.com/tracker and take the baseline described in Key 7 before you change anything.

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This guide is provided for general educational purposes and does not constitute individualized medical advice, diagnosis, or treatment. Always consult your physician before starting any new supplement or exercise program, particularly if you are pregnant, nursing, or taking prescription medications. Continue your GLP-1 medication exactly as prescribed by your physician.

References

Every citation below was independently retrieved and confirmed against its live PubMed record (esearch plus esummary or efetch) before use. None are drawn on trust from any prior document. Full verification detail, including claims investigated and left uncited, is in the companion file GLP1-Citation-Ledger.md.

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From Understanding to Doing

Everything up to this point was about understanding: what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Keep taking your GLP-1 medication exactly as your prescriber directs. Nothing in this guide is a reason to start, stop, adjust, or change the timing of a prescription. Semaglutide, tirzepatide, and liraglutide are prescription medications, and every decision about them belongs to the physician managing that part of your care. What this guide does is help you see the part of the picture that the prescribing panel was not designed to show.

The one emergency in this kit. If you have had persistent vomiting or months of very low intake on a GLP-1 medication, and you then develop confusion or unusual mental fuzziness, unsteadiness on your feet, or vision changes such as double vision or trouble moving your eyes, go to an emergency department the same day and say the words “possible thiamine deficiency.” That combination describes Wernicke’s encephalopathy. Treatment is intravenous thiamine and it needs to happen before the damage sets. This is rare and it is not a reason to stop your medication, and it is covered in full in Key 5 of *Unlock Your GLP-1*.

Anything severe or worsening goes to your prescriber, not to a supplement. New or escalating abdominal pain, persistent vomiting, symptoms that are getting worse rather than settling: those are clinical conversations, and a lab panel is not a substitute for one.

What standard GLP-1 monitoring is for, and what it was never built to see

If you are on a GLP-1 medication, you are almost certainly being monitored on some combination of HbA1c, fasting glucose, weight, lipids, and sometimes liver enzymes. That set is appropriate. It is the right way to manage the medication, it answers the questions your prescriber is responsible for answering, and I am not going to tell you it is worthless. It is doing its job.

It was simply built for a different job than the one this guide is about. Those markers track whether the drug is working. They were not designed to detect the cellular cost of mobilizing fat quickly, and they do not.

Here is the specific gap. A patient can walk in with a textbook HbA1c trajectory, falling weight, and improving lipids, and at the same time be running depleted glutathione, low CoQ10, and falling NAD+. Nothing on the standard panel reveals any of that, because nothing on the standard panel is looking for it. Key 1 and Key 3 of *Unlock Your GLP-1* describe the mechanism behind that gap: fatty acids released faster than mitochondria can process them, a surge of reactive oxygen species, and the senescent cells that follow (PMID 41920466, PMID 33262144). Hold that at the right level of confidence. The biology is laboratory work, and the step from it to how any specific person on a GLP-1 feels is a proposed mechanism rather than a proven chain. Key 1 says exactly that. I work from the model because it fits what I see and because everything it points to is low risk and defensible on its own terms.

So read this guide as an addition, not a correction. Your prescriber's panel stays. This is the second layer, and the markers below were chosen specifically to make the cellular picture visible.

The two jobs testing does here

The first job is deciding what you actually need. The antioxidant, mitochondrial, and NAD+ products marketed to GLP-1 patients are not cheap, and buying the full stack on the assumption that you are short in all of it is guesswork. The most expensive supplement is the one you did not need. Testing first is what turns a protocol from presumptive into directed, and it is the reason my own program reviews labs at the week two visit before the second layer of support is finalized.

The second job is documenting change against your own starting point. Ninety days that end with “I think I feel better” is a weaker result than ninety days that end with a documented shift in glutathione, GGT, and NAD+ next to your own intake numbers. I am not going to tell you how much any of them will move or by when. What a retest gives you is your own before and after, judged against your own log rather than against anyone’s promise, including mine.

Those are the two reasons to spend money on any of this. If a test cannot do either one for you, it is not worth running.

The blood sugar panel, read properly

This is the one place where the standard panel and this guide are looking at the same markers, and the difference is what gets ordered alongside them.

Ask for fasting glucose, HbA1c, and fasting insulin together. Fasting insulin is the one that routinely gets left off, and it is frequently the most informative of the three, because insulin climbs for years to hold glucose steady before glucose itself ever drifts. Measuring only glucose means finding the problem after the compensation has already failed, which is the wrong end of the window to be working in.

HOMA-IR is not a separate blood test. It is an index calculated from your fasting insulin and fasting glucose, which is why both have to be drawn at the same fasting draw for it to mean anything. I use it to track metabolic recovery over the course of a program. GLP-1 therapy should be pushing this number down. My optimal is below 2.0, and above 2.5 is where I act.

Where that action goes matters. A persistently elevated or rising HOMA-IR on GLP-1 therapy is information for the physician managing your prescription, and my response is to notify them, not to adjust anything myself. On my side the levers are the ones that were always mine: carbohydrate load and resistance training. Insulin resistance independently drives cellular senescence, which means an elevated value here compounds the mechanism in Key 1 rather than sitting harmlessly beside it.

What this panel cannot tell you. A normal HbA1c does not mean your cellular picture is clean. That is the entire premise of this guide, and it is worth saying plainly rather than implying it.

The oxidative panel: glutathione, CoQ10, and GGT

Three markers, and they are most useful read together rather than one at a time. Fasting is not required for this group on its own, though see the section on getting the draw right, because the fasting requirement of the metabolic panel governs when both are drawn at one visit.

Glutathione, whole blood. A direct measurement of your primary intracellular antioxidant, measured in red blood cells. The reactive oxygen species load described in Key 1 draws this down. A low result means the cell has lost its main line of defense. My functional target is at or above 900 mcmol/L, and I intervene below that. It is also the marker I most often pick for a retest, because it is responsive to a glutathione precursor such as NAC paired with direct liposomal glutathione, and six to eight weeks is the interval I retest it at rather than waiting the full ninety days. Whether yours moves, and by how much, is what the retest is for. I am not going to tell you in advance. If it is low, I also confirm magnesium adequacy, since magnesium is a required cofactor for glutathione synthesis and a shortfall there caps what the precursors can do.

CoQ10, plasma. Coenzyme Q10 is the primary electron carrier in the mitochondrial electron transport chain. GLP-1 therapy alters cholesterol synthesis and lipid flux through the mevalonate pathway, and the 2026 redox review of GLP-1 metabolic flux describes the downstream consequences for CoQ10 and glutathione peroxidase (PMID 41920466). Depletion here impairs ATP production and increases reactive oxygen species output, which is a self-reinforcing loop rather than a static deficiency. My functional target is at or above 0.5 mcg/mL.

GGT. Gamma-glutamyl transferase is the enzyme that cleaves glutathione to recycle its amino acid components. It is a validated, standard liver enzyme, and it is already on most comprehensive metabolic panels, which makes it the cheapest marker in this entire guide. Reading it as an indirect oxidative stress marker is inference rather than a validated use, and I want you to know which of those two things you are looking at. The inference is this: rising GGT reflects increased glutathione turnover under oxidative demand, so it tends to move before direct glutathione has fallen out of range. My functional target is 10 to 17 IU/L, and I act above 17.

Reading GGT and glutathione together

These two answer different questions. Glutathione reports your current reserve. GGT reports how hard your body is working to replenish it. Neither one alone tells you what the pair tells you.

GGT	Glutathione	What I read from it
Above 17	Below 900	Maximum oxidative burden. Reserves depleted and actively under stress. This is the highest priority for antioxidant support in the whole panel.
Above 17	At or above 900	Compensated oxidative stress. Keeping pace, but working hard to do it. The point of intervening here is to prevent depletion rather than to correct it.
17 or below	Below 900	Depletion without current acute oxidative activity. Standard antioxidant support, without urgency.
17 or below	At or above 900	Antioxidant status adequate. I hold the antioxidant layer unless symptom burden is high, and reassess at ninety days.

One honest note on thresholds. These are my clinical targets, not validated diagnostic cutoffs, and they are deliberately narrower than the standard laboratory range. When a result lands within ten to fifteen percent of a threshold, I let symptom burden tip the decision rather than treating the number as a wall. That is a judgment call, and I would rather say so than pretend the line is sharper than it is.

Albumin, the cheapest early warning in the panel

Albumin is a liver-produced protein that reflects overall protein nutritional status and tissue repair capacity. It is on every comprehensive metabolic panel already, it costs nothing extra to read, and in this population it is the earliest practical warning signal for inadequate protein intake.

That matters here more than in almost any other program I run. A systematic review of 35 randomized trials found the proportion of GLP-1 weight loss coming from muscle-related indices exceeded the benchmark expected from diet alone in two-thirds of trials (PMID 41996180), and a 24-month retrospective cohort measured the consequence directly in muscle mass, grip strength, and gait speed (PMID 40631351). Falling albumin during GLP-1 therapy is actionable well before any of that becomes obvious to you.

My functional target is 4.0 to 5.0 g/dL. Below 4.0 is where I stop and have a direct conversation about protein, confirm that the 1.6 g/kg daily minimum from Key 2 is actually being met on a normal day rather than a good one, and recheck at 45 days. *The GLP-1 Diet Guide* in this kit has the calculation and the food list to count against. Below 3.5 g/dL I flag for prescriber review rather than handling it myself.

What albumin cannot tell you. It is not a muscle measurement. It is a protein-status signal that moves early, which is a different and more modest thing. If you want to know what your muscle is doing, the answers are grip strength, a functional assessment, and what your training log says, none of which come out of a tube (PMID 42356514).

NAD+ testing, and the honest version of what a result changes

This is the marker with the most marketing attached to it, so it gets the most careful section.

What the panel is. A nine-marker profile measured by LC-MS/MS, including NAD+, NADH, NADP, NADPH, nicotinic acid, and nicotinamide. It is collected as a dried blood spot by fingerstick at home, which removes the laboratory visit entirely. The two values I actually read are the NAD+ absolute value and the NAD+ to NADH ratio.

Why it is in this guide at all. The mechanism is real and it is specific to what you are going through. SASP, the inflammatory signal released by senescent cells, induces CD38 expression in neighboring healthy cells, and CD38 degrades NAD+ (PMID 33199924). Senescent cells therefore actively drain the energy currency of the tissue around them. NAD+ is required for DNA repair, sirtuin activation, and mitochondrial function, which is why this shows up as fatigue and fog that do not track with your improving metabolic numbers.

My thresholds. At or above 20 mcmol/L is where I want to see NAD+. Below 20, or a NAD+ to NADH ratio below 3 to 1, is a finding. Below 15 mcmol/L is a more emphatic one.

What this test cannot do, stated plainly. There is no validated clinical reference range for NAD+ in the sense that there is one for glucose or albumin. This is a functional marker, suggestive rather than diagnostic, and a number outside my target does not confirm a deficiency state or a diagnosis. It is one input.

And here is where I part company with the market. A low NAD+ result is not, by itself, a prescription for an NAD+ precursor, and I want to be specific about why rather than leaving you with a shrug. The human NMN trials that exist are small and measure laboratory markers rather than outcomes anyone can feel or that predict health years out (PMID 36482258). Separately, a 2024 analysis linked terminal breakdown products of excess niacin to major adverse cardiac events across multiple cohorts, with laboratory work showing one of those metabolites drives blood vessel inflammation (PMID 38374343). That is an association plus a mechanism, and it is not proof that an NAD+ precursor harms anyone. But niacin, NMN, and related compounds clear through a shared pathway, and if you are on a GLP-1 you are, almost by definition, someone whose cardiovascular risk is under active management. An unresolved vascular question against a benefit that has not been demonstrated is a poor trade in your specific situation.

So what do I do with a low NAD+ instead. I read it first as a marker of senescent burden and oxidative load rather than as an automatic supplement order. That points me back to the levers with the better evidence behind them: the protein and resistance training in Key 2, the antioxidant work above, the senolytic sequencing in Key 4, and the clearance work in Key 6.

Where a precursor does and does not enter the picture. *Unlock Your GLP-1* declines to recommend routine NAD+ precursor supplementation, and it is right to. Nothing here contradicts that, and neither does *The GLP-1 Supplement Guide* in this kit, which is where the full argument lives. The short version, so the two documents read as one position rather than two. A precursor is never added on spec. It waits for a measured result that has actually crossed the line, it goes in behind the levers above rather than ahead of them, it carries the unresolved vascular question in the paragraph above, and it is stopped and rechecked at ninety days rather than renewed out of habit. If you have known cardiovascular disease, it is the item I am most willing to leave out entirely. That is a decision to make in a conversation, not one to make from a table. If you have already been sold an NMN product on the strength of a low result, that is worth a conversation rather than a panic.

Biological age testing, and what it is actually good for

Methylation-based biological age testing reports an estimated biological age, a rate of aging via DunedinPACE, telomere length, organ-system-specific ages, and immune cell composition. A DunedinPACE value above 1.0 indicates aging faster than one year per calendar year, and above 1.2 is where I treat it as a priority finding rather than a curiosity.

Be clear about what that number is. A biological age result is an estimate built from a model. It is not a measurement of your cells, and the available models disagree with each other. It never confirms anything about your health, and anyone presenting it to you as proof is going well past what it can carry.

What it is genuinely useful for is a comparison against itself. Run the same test at intake and again at ninety days and you have a consistent number on your own record, measured the same way twice. That is a real before and after, and in my experience it is the marker patients find most motivating, which is not nothing when the hard part of this is sticking with the protein and the training for three months.

There is no supplement trigger attached to it. No result on a biological age panel changes the protocol I build. It is an outcome marker, not a decision marker, and I would rather you knew that before you paid for it.

A companion panel from the same collection reports GlycA as a composite chronic inflammation marker, oxidative defense proxies including GPX1, NAD+ metabolism proxies, mitochondrial function markers, a lipid panel, amino acids, IGF-1, adiponectin, HbA1c, glucose, and environmental toxin load markers. If you are choosing only one, the split is straightforward: the biological age panel is the stronger motivational marker and the better before-and-after, and the metabolic panel is the more clinically actionable of the two for protocol decisions. Pick based on which of those two purposes you actually want.

The nutrient panel, because you are eating substantially less food

This is the quietest section in the guide and it is the one I would defend hardest.

In the trial that worked out how semaglutide actually produces weight loss, total energy intake across a day fell by 24 percent (PMID 28266779). Your requirement for vitamins and minerals did not fall by 24 percent. Reviews of GLP-1 therapy now describe a recognizable pattern of micronutrient shortfalls following from sustained low intake, unevenly distributed, so not everyone is short in the same places (PMID 41549912). The literature is openly asking whether GLP-1 care should borrow the monitoring framework that bariatric surgery already has rather than reinvent one (PMID 41373949, PMID 29693218).

Thiamine, vitamin B1, deserves its own line. Your body holds a small reserve relative to daily use, so thiamine status falls within weeks of poor intake rather than over months. That timeline lines up uncomfortably well with GLP-1 therapy, particularly if nausea or vomiting has been part of your experience. In one obesity medicine clinic population, about a third of patients already tested thiamine deficient before any surgery or medication (PMID 39837759), and low thiamine is common in type 2 diabetes as well (PMID 17676306). Thiamine is measurable, which puts it firmly in the test-rather-than-guess category. The safety picture behind this is in the red flag at the top of this guide, built on a systematic review of published cases with semaglutide (PMID 42399213) and a pharmacovigilance analysis across GLP-1 medications (PMID 41534460). Rare, real, and worth recognizing early.

B12 and RBC folate. Ask for red blood cell folate rather than serum folate, because it reflects roughly three months of tissue status rather than this week's meals. Long-term acid suppression and metformin both reduce B12 absorption, and a good number of GLP-1 patients are on one or both, so if nobody has checked, ask.

Homocysteine, read correctly. Homocysteine is a methylation-cycle marker: it rises when B12, folate, or B6 run short, and with thyroid or kidney trouble. Do not let anyone wave off an elevated result as trivia. In my read this is one of the most consequential numbers in the panel, because it tells you the methylation machinery is running short somewhere findable. The way to act on it is not to chase the number with blind supplementation. Find which input is short, B12, folate, B6, thyroid, or kidney function, correct that, and let homocysteine confirm the fix on retest. My functional target is 5 to 7. And lower is not better: a homocysteine driven too low is its own finding, because it can signal overmethylation and B vitamin intolerance rather than health. When B vitamin repletion turns out to be the fix, the combination I use is B12 at 2000 mcg alongside folate, B6, and TMG, which is trimethylglycine. Homocysteine Supreme is the closest single product to that combination.

Vitamin D. Worth testing and worth correcting a genuine shortfall. My functional target is 40 to 80 ng/mL, and below 40 ng/mL is where I act. Below 20 ng/mL I want the prescribing physician involved rather than handling it as a supplement question.

Ferritin. Rapid weight loss on sharply reduced intake is a reasonable time to look at iron status, and ferritin is where I start. It rises with inflammation and can look reassuring in someone genuinely depleted, which is one of two reasons the hs-CRP below earns its place.

hs-CRP. A nonspecific screening marker. A raised result says an inflammatory load exists somewhere. It does not say where it is coming from, and it never confirms a diagnosis. Here it does two jobs: it is the sanity check on your ferritin, and it is a rough read on the systemic inflammatory picture that the senescence model in Key 3 predicts.

Functional ranges for the markers in this guide

Clinical targets I use at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose. Where I have written that a standard range varies by method, that is because there is no single agreed one to print, and I would rather say so than invent a number.

Marker	Units	Functional target	Approximate standard range
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
Fasting insulin	uIU/mL	2.6 to 5.0	2.0 to 10.0
HOMA-IR	index	below 2.0	calculated, not reported by the lab
Glutathione (whole blood)	mcmol/L	at or above 900	varies by laboratory method
CoQ10 (plasma)	mcg/mL	at or above 0.5	varies by laboratory method
GGT	IU/L	10 to 17	up to 55
NAD+	mcmol/L	at or above 20	no validated clinical range
NAD+ to NADH ratio	ratio	at or above 3 to 1	no validated clinical range
Albumin	g/dL	4.0 to 5.0	3.5 to 5.0
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Homocysteine	umol/L	5 to 7	under 10 to 15, varies by lab
Ferritin	ng/mL	50 to 150	25 to 300
hs-CRP	mg/L	under 0.5	under 1.0
DunedinPACE	ratio	below 1.0	1.0 equals one year of aging per calendar year

Thiamine is deliberately absent from this table. I recommend testing it and I have said why, but I only publish a functional target I can transcribe exactly, and the assay units and cutoffs vary enough between laboratories that printing one here would be a guess dressed up as a range. Ask your laboratory which assay it runs and read your result against its own reference.

Reading your results as a pattern

Pattern one: perfect metabolic panel, depleted cellular panel. HbA1c falling, lipids improving, weight down, and glutathione below target with GGT above it. This is the picture this entire guide was written for, and it is the most common one I see. Nothing is wrong with the medication. The cellular layer simply was not being watched.

Pattern two: albumin drifting down. Often the first thing to move, and often in someone who genuinely believes they are eating enough protein. Count a normal day honestly against the 1.6 g/kg minimum before concluding anything.

Pattern three: HOMA-IR not falling. GLP-1 therapy should be pushing this down. If it is flat or climbing, that is a conversation with the prescribing physician plus a hard look at carbohydrate load and resistance training, in that order.

Pattern four: everything in range. This happens, and it is a real result rather than a disappointment. It means the antioxidant and mitochondrial layer is not where your money should go, and the protein, the training, and the clearance work from Keys 2 and 6 carry the program. Hold and reassess at ninety days.

Pattern five: one marker badly out, everything else clean. Chase the one. A single low CoQ10 with adequate glutathione and normal GGT is a narrower problem than a whole depleted panel, and it should get a narrower response.

What these labs cannot tell you

- **Whether your medication is causing your symptoms.** No marker in this guide attributes anything to your prescription. It documents a state, not a cause, and the causal chain from that state to how you feel is a proposed mechanism rather than a proven one. Key 1 of the ebook says the same thing and I am repeating it here on purpose.
- **Whether you have lost muscle.** Albumin warns early about protein status. It does not measure lean mass. Grip strength, a functional assessment, and what you can actually lift answer that question.
- **Whether your mitochondria are working.** CoQ10 and NAD+ read on inputs to mitochondrial function. There is no validated clinical test of mitochondrial function itself, and anyone selling one is ahead of the evidence. So I do the two things that are actually available. I read the inputs I can measure, CoQ10 and NAD+, and I judge function by what reflects it in daily life: training tolerance, how you feel in the two days after a session, and the energy score in your log. That is a cruder instrument than a validated assay, and it is the honest one on offer.
- **How much your biological age will move.** It is a model output, run twice for comparison. It is not a forecast and I will not present it as one.
- **Whether a supplement is working.** Only a retest of the same marker answers that, which is the entire argument for taking a baseline before you change anything.

What I would not build a plan on

- **A single NAD+ value in isolation**, particularly as the justification for buying an NAD+ precursor. See the section above for why.
- **A biological age result on its own.** One number from one model, with no second run to compare it to, is a conversation piece.
- **Any panel whose interpretive framework exists only in the marketing material of the company selling it.** That test is being asked to do a job it was never validated for.
- **A retest that added new markers.** Adding markers at the ninety day draw produces data with no baseline to compare against, which is the opposite of the point. Retest what you drew at intake.
- **Symptoms alone, with no baseline.** Take the baseline in Key 7 before you change anything: energy out of ten, sleep out of ten, and how you feel in the two days after a training session. Then judge everything against your own log.

Getting the draw right

Small logistics, and they decide whether the results are usable.

The fasting question, which trips people up. The metabolic markers require fasting and the oxidative markers do not. When both are drawn at one visit, the combined requisition inherits the fasting requirement. Fast from midnight and schedule a morning appointment. No food, coffee, juice, or supplements for eight to twelve hours beforehand. Water is fine. If you arrive non-fasting, the oxidative markers can still be drawn, but the insulin result will be invalid and you will need a second visit, which is entirely avoidable.

At the patient service center. Bring a printed paper copy of your requisition, since a phone screen will not do, plus a photo ID and your insurance card if you have one. Tell the phlebotomist it is a fasting draw. You can find your nearest LabCorp location at labcorp.com or by calling 1-800-522-7096, and while most locations accept walk-ins, scheduling online is easier. Results reach me in two to five business days, and I review them with you at your week two call.

Home fingerstick kits. The NAD+ profile and the biological age panels are collected at home on dried blood spot cards, which removes the laboratory visit and the draw fee entirely. Three things matter. Collect in the morning and follow the kit instructions exactly, since the whole process takes about ten minutes. Register the kit online with the enclosed code before mailing it, because unregistered kits cannot be processed. And mail it back in the prepaid envelope the same day you collect, without exception. NAD+ results take approximately fourteen business days, and biological age results take approximately three to four weeks.

The retest, and when to order it. Order the ninety day retest at week ten to eleven so the results are in hand at the ninety day visit rather than arriving after it. A comparison that lands after you have mentally closed the program does much less for you than one you can look at together.

When to seek clinical review

Same day, emergency: confusion, unsteadiness, or vision changes after persistent vomiting or a long stretch of very low intake. Say “possible thiamine deficiency” at the emergency department.

Promptly, with your prescriber: new or worsening abdominal pain, persistent vomiting, an albumin below 3.5 g/dL, a vitamin D below 20 ng/mL, or a HOMA-IR that is flat or rising rather than falling on GLP-1 therapy.

At your next appointment: any of the cellular markers out of my functional range, symptoms from Key 3 that are new since starting your medication, or a supplement protocol you were sold without a lab result behind it.

Track what changes, from anywhere

Take the baseline before you change anything. Energy, sleep, and how you feel two days after training, plus your protein on a normal day. Labs tell you about the tissue. Your log tells you about you, and the ninety day comparison needs both halves.

Root Tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=glp1-labs

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Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Every panel and every question in this guide works just as well in your own physician's office, and I would rather you ran them there than not at all.

If you would rather start on your own, start your free tracker at rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=glp1-labs and take the baseline before you change anything.

You can also reach the office at (919) 662-0520.

Disclosure: where laboratory testing is arranged through my practice, Root Health earns a portion of what you pay for it. Nothing in this guide was included, excluded, or thresholded based on what it earns, and the most heavily marketed test in this space, the NAD+ profile, is the one this guide tells you not to act on by itself. You are free to obtain any of these tests through your own physician instead.

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Every citation below is drawn from the verified citation record for this program, `GLP1-Citation-Ledger.md`, where each was confirmed against its live PubMed record before use.

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For educational purposes only. This guide does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship.

Root Health | Dr. Adam Seay, DC | Garner, NC | rootcauseunlocked.com

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. Nothing here is a reason to start, stop, or change a prescription on your own. Keep taking your GLP-1 medication exactly as your prescriber directs, and take every question about dose, timing, escalation, or side effects to the physician managing that part of your care. That is their call, not mine, and not something to work out from a document.

One emergency belongs at the top of this guide rather than buried in it. If you have had persistent vomiting or months of very low intake, and you then develop confusion or unusual mental fuzziness, unsteadiness on your feet, or vision changes such as double vision or trouble moving your eyes, go to an emergency department the same day and say the words "possible thiamine deficiency." That combination describes Wernicke's encephalopathy. The treatment is intravenous thiamine and it needs to happen before the damage sets.

Keep it in proportion. A systematic review of published cases found six reported with semaglutide (PMID 42399213), and an analysis of FDA adverse event reports identified fifteen across GLP-1 medications as a whole (PMID 41534460). Set against many millions of prescriptions, that is a very small number, and it is not a reason to stop your medication. Two details from those reports are worth carrying with you anyway: nearly every case had a warning period of prolonged gastrointestinal symptoms or rapid weight loss first, and lasting neurological problems were common among patients followed afterward. Caught early it is treatable. That is the entire reason it is on page one of a diet guide.

Seek care promptly, short of an emergency, if you cannot keep fluids down, if you are lightheaded on standing, or if you have stopped urinating normally. Dehydration on a suppressed appetite arrives quietly.

Talk to me or your physician before you take on the protein target in this guide if you are pregnant or nursing, have kidney disease, have a history of disordered eating, or are managing diabetes on medication that can drop your blood sugar. A high protein target is not automatically right for every kidney, and reduced eating plus glucose-lowering medication is a combination worth supervising rather than improvising.

What food can and cannot do here

Your supplements are doing real work. So is your training. But food is the foundation underneath both of them, and no supplement stack fully overcomes a diet that is working against it.

Here is the situation your body is actually in. Your GLP-1 medication reduces your appetite significantly, which is exactly what it is designed to do. The problem is that eating less means taking in less of everything: fewer calories, but also less protein, fewer vitamins and minerals, and fewer of the plant compounds that protect your cells. For most people, the first thing to disappear when appetite drops is not junk food. It is protein. And the tissue that pays the price is muscle.

What is established, and I will name the evidence rather than gesture at it.

- **Intake falls hard.** In the trial that worked out how semaglutide actually produces weight loss, total energy intake across a day fell by 24 percent (PMID 28266779). Your requirement for protein, vitamins, and minerals did not fall by 24 percent alongside it.
- **The weight coming off is not all fat.** A systematic review of 35 randomized trials comparing incretin-based and non-drug weight loss found the median share of weight lost as muscle-related mass was 28.3 percent on incretin therapy, with two thirds of the trials landing above the roughly 25 percent benchmark expected from weight loss alone (PMID 41996180).
- **Low protein intake is one of the named drivers.** A review of the mechanisms behind GLP-1 associated muscle loss lists reduced protein intake alongside negative energy balance, hormonal shifts, impaired muscle-building signaling, and increased protein breakdown (PMID 41754149). Food is the lever you personally control in that list.
- **The countermeasure is not exotic.** Protein intake paired with resistance training is the most direct, best supported response to this specific problem (PMID 42356514). The consequences of not responding are measurable: reduced muscle mass, grip strength, and gait speed have been documented in GLP-1 treated older adults over 24 months (PMID 40631351).
- **Micronutrients follow calories down.** Reviews of GLP-1 therapy now describe a recognizable pattern of micronutrient shortfalls that follow from sustained low intake, unevenly distributed from person to person (PMID 41549912).

What is reasoning, and I am labeling it as such. The second half of this guide is a list of foods chosen for their effect on cellular inflammation. The underlying biology is sound: senescent cells stop dividing without dying and release a continuous inflammatory signal, which is well established laboratory work (PMID 33262144). What has not been tested is whether eating these particular foods changes how a person on a GLP-1 medication feels or functions. Nobody has funded that trial, which is what happens to compounds nobody can patent. So the food list is reasoned clinical practice held to an endpoint, not trial-proven certainty, and I would rather you know which one you are getting.

What food cannot do here. It cannot replace resistance training, because protein without a stimulus to build has nothing to build toward. It cannot fix a protein target you never reach. And it cannot substitute for the conversation with your prescriber if nausea has made eating genuinely impossible.

The next section is the single most important thing in this guide. Everything after it builds on top of it, so start there and hold onto it even in a week when nothing else holds.

The single most important habit in this guide

Eat your protein first at every meal, before vegetables, grains, or anything else.

Your appetite will run out mid-meal. When it does, you want the protein already eaten. That is the whole idea. It costs nothing, it requires no product, and it survives a bad week better than any meal plan.

Your daily protein target

Aim for **1.6 to 2.0 grams of protein per kilogram of your body weight every day**. This is higher than standard dietary guidance, and deliberately so. During rapid weight loss your body breaks down muscle tissue to harvest amino acids, and the only way to counteract that is a steady supply of dietary protein.

Calculate your target

- My weight: ____ lbs
- Divided by 2.2 = ____ kg
- Multiplied by 1.6 = ____ grams of protein per day (your minimum)

Example: 180 lbs divided by 2.2 equals 82 kg. 82 times 1.6 equals 131 grams daily.

That is the floor, not the ceiling. The upper end of the range, 2.0 grams per kilogram, is where I would sit someone who is training hard and losing weight quickly, assuming their kidneys are normal.

Spread it across the day

Your muscles respond best to **30 to 45 grams of protein at a time**. Three meals at that level beats one large protein meal, even if the daily total is identical. Here is a practical way to think about your day.

Meal	Protein target	What that looks like
Breakfast	30 to 40 grams	Two eggs plus a cup of Greek yogurt. Or one cup of cottage cheese with berries. Or a protein shake made with real food added.
Lunch	35 to 45 grams	A four to five ounce serving of chicken, fish, or beef as the centerpiece, with vegetables alongside rather than in place of it.
Dinner	35 to 45 grams	Same principle. Animal protein leads. Everything else supports it.
After exercise	25 to 40 grams	Within two hours of finishing a resistance session. This window matters and skipping it wastes the work you just did.

Key 2 of *Unlock Your GLP-1* covers the training side of this in more detail. Protein and resistance training are two halves of one intervention, and neither half works properly alone.

Best protein sources

Animal proteins are richer in leucine, the amino acid that actually triggers muscle building, which is why they lead this list. Plant proteins are included and are useful, but they should supplement rather than replace animal sources during this program unless you have a dietary reason to avoid them.

Food	Serving	Protein
Chicken breast or thigh	4 oz cooked	28 to 32 grams
Lean ground beef, 90 percent or leaner	4 oz cooked	26 to 28 grams
Wild caught salmon	4 oz cooked	25 to 28 grams
Cottage cheese	1 cup	25 to 28 grams
Canned sardines	1 can, 3.75 oz	22 to 24 grams
Canned tuna in water	3 oz	20 to 22 grams
Greek yogurt, plain	1 cup	15 to 20 grams
Lentils	1 cup cooked	18 grams
Edamame	1 cup	17 grams
Eggs	2 large	12 to 13 grams

If you are struggling to eat enough

Nausea and early fullness are common on GLP-1 medications. A few things that help:

- **Eat smaller amounts more often** rather than three large meals. Four or five smaller protein-forward meals often work better than three.
- **Cold foods are frequently easier to tolerate** than hot foods when nausea is present. Cottage cheese, Greek yogurt, and cold sliced chicken are useful here.
- **Liquid protein counts.** A protein shake is a legitimate meal when solid food is difficult.
- **Do not drink large volumes of fluid with meals.** It fills the space you need for food. Hydrate between meals instead.

If you consistently cannot reach your protein target, tell me. This is a clinical problem worth solving, not something to push through alone. And if the barrier is the medication itself, that conversation belongs with your prescriber, because adjusting a GLP-1 dose is their decision and not something I would advise from here.

The nutrient side of eating less

Protein gets the attention because muscle is visible. The quieter problem is everything else that falls with intake.

Thiamine, vitamin B1, deserves singling out, because it behaves differently from the rest. Your body holds only a small reserve relative to what it uses daily, so thiamine status falls within weeks of poor intake rather than over months. In one obesity medicine clinic population, about a third of patients already tested low before any surgery or medication (PMID 39837759). Bariatric surgery creates the same combination of sharply reduced intake plus vomiting, and precisely because of that it has an established thiamine monitoring protocol built around it (PMID 29693218). Reviews are now asking directly whether GLP-1 care should borrow that framework rather than reinvent it (PMID 41373949).

How I handle this clinically. I test rather than guess, because the most expensive supplement is the one you did not need. Food comes first: the protein sources above carry B vitamins, zinc, and iron along with the protein, which is one more reason a shake alone is a poor long-term substitute for a meal. Where intake has been low for a sustained stretch, or where nausea and vomiting have been a recurring part of your experience, that is a reason to look at nutrient status directly rather than assume it is fine. *Understanding Your GLP-1 Labs* in this kit covers which markers are worth drawing, including thiamine, and what each one can and cannot tell you. Clear any new supplement with your prescriber first, particularly if you take a blood thinner, thyroid medication, or a diuretic.

Foods that work against cellular inflammation

The cellular aging process this program addresses is fundamentally an inflammatory one. Certain foods work against it. These are not folk remedies. The compounds in them are the same ones appearing in pharmaceutical research on cellular aging.

Before the list, the honest caveat, because it changes how you should read every item below. In a screening of candidate compounds, fisetin was identified as the most potent natural senolytic tested (PMID 30279143), and that research is real. But senolytic research uses concentrated compounds on short pulsed schedules, which is how the mechanism was studied (PMID 40994903). A cup of berries is not a senolytic dose and I am not going to pretend otherwise.

So why is the list here at all? Because every food on it earns its place twice over: it is a whole food with a low refined-carbohydrate load, it fits inside a small appetite without wasting the space, and it carries the compounds the research is built on even if it does not carry the research dose. That is the claim I can stand behind. Anyone selling you the stronger version is ahead of the evidence.

Berries

Blueberries, blackberries, and strawberries contain fisetin and quercetin, the same natural compounds used in senolytic research to clear senescent cells. Aim for **at least one cup daily**. Frozen berries work as well as fresh for this purpose and usually cost less, so buy frozen without hesitation.

Onions, capers, and apples

These are among the highest dietary quercetin sources available. Red onions carry roughly twice the quercetin of white. Apple skin holds most of the apple's quercetin, so eat the whole fruit rather than peeling it.

Fatty fish

Salmon, sardines, and mackerel supply EPA and DHA, the omega-3 fatty acids that reduce inflammatory signaling. **Aim for fatty fish at least three times per week.**

This is the one item on the list with direct human trial evidence behind it, and I want to be precise about its size. A meta-analysis of randomized trials found omega-3 supplementation produced a very small but statistically significant benefit to muscle strength, with no meaningful effect on muscle mass or function overall (PMID 36811583). The most encouraging single trial gave healthy adults aged 60 to 85 about 3.4 grams of combined EPA and DHA daily for six months and found gains in thigh muscle volume, grip strength, and one-rep-max strength against control (PMID 25994567). Note the dose. That is more than a standard fish oil capsule and more than most weeks of eating deliver. Omega-3 is a garnish, not the meal. Key 2 of *Unlock Your GLP-1* covers the reasoning, and *The GLP-1 Supplement Guide* in this kit carries the dose detail if you want the full picture.

Green tea

Contains EGCG, a compound studied for its effect on the inflammatory output of senescent cells. **One to two cups daily.** Decaffeinated green tea retains most of the EGCG if caffeine is a problem for you.

Extra virgin olive oil

Contains oleocanthal, which laboratory work indicates inhibits the same inflammatory enzyme targeted by ibuprofen. That peppery burn at the back of your throat from good olive oil is the oleocanthal. I am not attaching a citation to that comparison because the work behind it is enzyme chemistry rather than a trial in people, and you should read it as a reason olive oil belongs in your kitchen rather than as a claim about pain relief. Use it generously as your primary cooking and finishing oil rather than sparingly.

Turmeric with black pepper

Curcumin acts on the inflammatory signaling that drives cellular aging, which is well documented in the laboratory and largely untested as a food intervention in people like you. Absorption from food is poor unless it is consumed with black pepper and a fat source together, so add turmeric to olive oil based dishes and always include pepper.

Fermented foods

Plain kefir, plain yogurt, kimchi, and sauerkraut supply live bacterial cultures. Your gut lining matters here more than most people realize. The proposed mechanism is that when the barrier becomes permeable, bacterial products enter circulation and activate the same inflammatory pathway that drives cellular aging. Fermented foods daily are a cheap, low-risk way to support that barrier, and the plain versions carry protein worth counting.

What to reduce

These foods activate the same inflammatory pathway this program is working to suppress. You do not need to eliminate them perfectly. You need to reduce them consistently.

Reduce	Why
Refined sugar and high fructose corn syrup	Activates the inflammatory signaling that drives cellular aging. It also binds to proteins including collagen, which is the proposed mechanism behind the structural breakdown in skin and joints this program is working against.
Corn, soybean, sunflower, and canola oils	Very high in omega-6 fatty acids, which worsens your omega-6 to omega-3 balance and increases the type of oxidative damage this program is addressing.
Ultra processed and packaged foods	Industrial additives and emulsifiers are associated with inflammatory pathway activation. These foods also crowd out the protein and whole food compounds you actually need, which matters more than usual when your appetite is limited.
Charred and processed meats	Charring produces inflammatory compounds. Deli meats and hot dogs are associated with elevated markers of cellular aging, which is an association rather than a demonstrated cause.
Alcohol	Increases gut permeability, allowing bacterial products into circulation where they trigger inflammation. It also depletes glutathione, the antioxidant this program is specifically working to rebuild.
Sweetened drinks	They bypass fullness signals entirely and displace food you need. With appetite already suppressed, liquid calories should be water, green tea, or a protein shake.

Hydration

Eight to ten glasses of filtered water daily. Drink between meals rather than with them, so fluid does not take up the stomach space you need for food.

If your food intake has dropped significantly, add an unsweetened electrolyte powder once daily, since reduced eating also means reduced sodium, potassium, and magnesium intake from food. This is one of the most common things I correct in people who feel lightheaded and assume it is the medication.

The overall pattern

What you are building toward is essentially a Mediterranean pattern with the protein turned up. Animal protein or legumes at the center of every meal. A wide variety of colorful vegetables. Extra virgin olive oil as your primary fat. Fatty fish several times a week. Very little refined carbohydrate. Fermented food daily.

I am going to label this honestly. The Mediterranean pattern is one of the best studied ways of eating there is, and it consistently performs well on inflammatory markers. Those trials were not run in people on GLP-1 medications, so what you are getting from this pattern is strong general evidence applied to your situation by reasoning, plus one part that is specific to you and well supported: the protein.

Your shopping list

Print this page or photograph it and take it to the store. Buying the right things is most of the battle.

Protein

Eggs · Chicken breast and thigh · Wild caught salmon · Canned sardines · Canned tuna in water · Lean ground beef · Plain Greek yogurt · Cottage cheese · Plain kefir · Lentils · Edamame · Whey or collagen protein powder

Produce

Blueberries, frozen or fresh · Blackberries · Strawberries · Red onions · Garlic · Leeks · Asparagus · Spinach · Kale · Arugula · Broccoli · Apples with skin · Avocado · Lemons

Pantry and other

Extra virgin olive oil · Ground turmeric · Black peppercorns · Green tea · Walnuts · Capers · Sauerkraut · Kimchi · Chia seeds · Quinoa · Steel cut oats · Sea salt · Electrolyte powder, unsweetened

A simple day

You do not need recipes. You need a pattern.

Meal	Example
Breakfast	Two eggs cooked in olive oil with spinach, plus a cup of plain Greek yogurt with half a cup of blueberries. Green tea.
Lunch	Four to five ounces of grilled chicken over arugula with red onion, olive oil, and lemon. Whole apple.
Dinner	Four to five ounces of salmon with roasted broccoli and asparagus in olive oil, seasoned with turmeric and black pepper. Small side of sauerkraut.
If needed	Cottage cheese, a handful of walnuts, or a protein shake. Use these to close a protein gap rather than as a habit.

Running a fair experiment

You are going to want to know whether any of this is doing anything. Here is how to find out in a way that can actually give you an answer.

Take a baseline before you change anything. Energy out of ten. Sleep out of ten. How you feel in the two days after a training session. Grip strength if you have any way to measure it, and how many repetitions you get on one movement you can repeat identically in a month.

Change the protein first and change it alone. It is the piece with real evidence behind it, it is measurable, and if you stack six changes at once you will never know which one mattered.

Count your protein for one honest week. Not forever. In my experience most people are eating well short of what they assume they are, and a single honest week of counting is usually enough to show you where the real gap is.

Give it a realistic window. Muscle responds over months, not over a couple of weeks. Judge this across the full ninety days against your own log, not against anyone's promise, including mine.

Judge it on function, not on the scale. The scale is already moving for reasons that have nothing to do with this guide. Strength, stairs, and how you feel two days after training are the honest readouts.

If ninety days of your own log shows nothing moved, that is a real result. Bring it to me and I will look at why with you rather than adding more to the pile.

Track it: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=glp1-diet

Progress, not perfection

You will not do this perfectly, and you do not need to. Every meal where protein comes first is a win. Every week with three servings of fatty fish is a win. Every day you skip the sweetened drink is a win. Consistency across ninety days beats perfection across five days and then quitting.

When to seek clinical review

Urgently: confusion, unsteadiness, or vision changes after a stretch of persistent vomiting or very low intake, which is the thiamine emergency described at the top of this guide. Also inability to keep fluids down, or lightheadedness that does not resolve when you sit down.

Before increasing your protein target: if you have kidney disease, are pregnant or nursing, or have a history of disordered eating.

With your prescriber, not with me: anything about your GLP-1 dose, timing, escalation, or whether nausea warrants a change. I do not prescribe these medications and I do not advise on adjusting them.

At your next appointment: if you consistently cannot reach your protein target, if you are losing strength rather than just weight, or if you have been on a suppressed appetite for months without anyone checking your nutrient status.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=glp1-diet

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you are eating as well as you can and still losing strength, that is worth a conversation rather than another rule about food. You can also reach the office at (919) 662-0520.

Not ready for a conversation yet? Start tracking: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=glp1-diet

Everything in this guide also works without me, and I would rather you use it either way.

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Keep taking your GLP-1 medication exactly as your prescriber directs. Semaglutide, tirzepatide, and liraglutide are prescription medications. Nothing in this guide starts one, stops one, adjusts a dose, or delays an escalation. If anything here raises a question about your prescription, that question goes to the physician managing it, not to me and not to a supplement.

The one emergency in this guide. If you have had persistent vomiting or months of very low intake on a GLP-1 medication, and you then develop confusion or unusual mental fuzziness, unsteadiness on your feet, or vision changes such as double vision or trouble moving your eyes, go to an emergency department the same day and say the words “possible thiamine deficiency.” That combination describes Wernicke’s encephalopathy. Treatment is intravenous thiamine and it needs to happen before the damage sets. This is rare, and it is not a reason to stop your medication. It is a reason to know the pattern.

This protocol is not appropriate in pregnancy or while nursing. If you become pregnant while running it, stop the enhanced layer immediately and contact both me and your obstetric provider.

If you take an anticoagulant or antiplatelet medication, two products in this protocol need to be cleared with your prescriber before you start, and one of them may be excluded outright. The interactions section covers which, and why.

If you take levothyroxine, a quinolone or tetracycline antibiotic, or a thiazide diuretic, the timing and dosing details in this guide are not optional decoration. Read the interactions section before your first dose.

Supplements in the United States are regulated as food, not as drugs. Quality varies and label accuracy is not automatic. Look for independent third-party testing.

Why any of this exists

Unlock Your GLP-1 covers the mechanism in full, and this guide assumes you have read it. The short version, because the protocol below makes no sense without it.

GLP-1 receptors are not confined to the pancreas and the appetite centers of the brain. They are also present on adipose-derived stem cells and on the fibroblasts in your skin.[1] A controlled study measuring stem-cell populations in people taking GLP-1 receptor agonists found real differences against controls.[2] At the same time, fat coming apart quickly releases fatty acids faster than your mitochondria can process them, and separate 2026 work describes the redox consequences of that flux, including downstream effects on CoQ10 and glutathione peroxidase.[3] Cells that take on too much of that oxidative damage stop dividing but do not die, and instead release a continuous inflammatory signal called SASP.[4]

This runs alongside your medication, not against it. The receptor your medication activates has documented protective effects of its own, including promotion of autophagy and suppression of inflammatory signaling.[5] The problem this protocol addresses is a separate, tissue-specific consequence of how fast the fat is coming off. Both are happening at once, in different tissues. Nothing below is designed to blunt the benefit you are already getting.

What you are buying here, stated plainly

I would rather say this once, clearly, than let you discover it later.

The senescence and SASP biology is well established in the laboratory. The step from that biology to how a specific person on a GLP-1 feels is a proposed mechanism, not a proven chain. **No human outcome trial has tested this protocol, or anything resembling it, in people on a GLP-1 medication.** Nobody has funded one. That is what happens to compounds nobody can patent: the trial never gets paid for, and marketing fills the space where the evidence should be.

So what this document contains is laboratory evidence, mechanism, and my own clinical use of it, held to a defined endpoint and a 90 day retest rather than run indefinitely.

Two things follow, and both are practical.

First, the protein and the resistance training come before anything you swallow. Those are the two levers in this whole field with real human trial evidence behind them, and they are the next section for that reason.

Second, everything else answers to the retest. If ninety days of your own numbers and your own log show nothing moved, that is a real result, and the right response is to stop rather than to renew.

The two levers that outrank everything on the shelves

GLP-1 weight loss is not entirely fat loss. A systematic review of 35 randomized trials found the median proportion of weight loss coming from muscle-related indices was 28.3 percent with incretin therapy, with two thirds of studies exceeding the roughly 25 percent benchmark expected from weight loss generally.[6] The mechanism is not one thing. Reviews describe reduced protein intake driven by appetite suppression, negative energy balance, hormonal shifts, impaired anabolic signaling, and increased protein breakdown working together.[7] A 24-month retrospective cohort measuring muscle mass, grip strength, and gait speed found accelerated sarcopenia in older adults with type 2 diabetes on semaglutide.[8]

The countermeasure is protein and resistance training, and the literature says so directly.[9]

- **Protein: 1.6 to 2.0 grams per kilogram of body weight per day**, distributed across three to four meals rather than back-loaded into dinner. Appetite suppression makes this genuinely hard, which is why protein goes on the plate first, before anything else. *The GLP-1 Diet Guide* in this kit has the calculation, the per-meal split, and the food list.
- **Resistance training a minimum of three times per week.** Compound movements, meaning squats, rows, presses, and hip hinges, worked to within two to three repetitions of failure, with a protein-rich meal or shake within two hours afterward.

If your budget, your appetite, or your week is finite, and everyone's is, this is where it goes first. Every product below is built to support that work, not to substitute for it.

How this protocol is built: two layers

The protocol has a core layer and an enhanced layer, and the difference between them is the most useful thing in this document.

The core layer is what I use for every person on a GLP-1, regardless of what their labs show, because the mechanisms it addresses are present in everyone losing fat this fast.

The enhanced layer is added only when a laboratory result or a documented symptom pattern confirms the specific thing that product addresses. Nothing in the enhanced layer is added before the result that justifies it exists.

This is deliberate, and it does three things.

1. **It cuts the daily count.** A full unfiltered version of this approach runs to fifteen or sixteen products a day. The two-layer build brings the typical day to eight to ten. That matters more than it sounds when nausea and a suppressed appetite are already part of your life and swallowing things is the problem.
2. **It keeps you from carrying products your biology does not require.** The most expensive supplement is the one you did not need.
3. **It keeps the second layer honest.** A product that gets added only after a number crosses a line is a product you can also stop when that number comes back across it.

What the layers usually add up to.

Where you are starting from	Core	Enhanced	Typical daily total
Lower burden. Intake questionnaire score 0 to 11, laboratory values within optimal ranges.	6	0 to 1	6 to 7
Moderate burden. Score 12 to 23, one or two confirmed deficiencies.	6	2 to 4	8 to 10
Higher burden. Score 24 to 36, multiple confirmed deficiencies and high symptom burden.	6	4 to 6	10 to 12

The core count holds at six because the drainage sequence contributes exactly one remedy at a time, run in order across the twelve weeks.

If you are holding this guide without the program

Some readers buy this guide on its own, without the structured program, the intake questionnaire, or the calls that come with it. It still works, and here is how to run it.

The core layer needs no laboratory results and no call with me. Everything in it is there because the mechanism it addresses is present in everyone losing fat this fast. Start it as written, including the pulse calendar and the drainage sequence.

The enhanced layer waits for laboratory results either way. Take the marker table in this guide, along with *Understanding Your GLP-1 Labs* from this kit, to your own physician and ask for the draws. The thresholds printed here tell you both exactly what I would act on. Where a trigger refers to the intake questionnaire or a call you do not have, let your laboratory results and an honest read of your symptom burden with your own physician make the call instead.

And if you want my read on your numbers, that conversation is free. The Root Discovery Session at the end of this guide is the door.

Layer one: the core

Product	Dose	When
Senolytic Synergy (Designs for Health)	2 capsules	3 consecutive days per week. Not daily. See the pulse calendar below.
Orthomega 820 (Orthomolecular)	Per label direction	With the largest meal of the day.
Magnesium Bisglycinate (DFH)	Per label direction	Once daily in the evening. Minimum two hours separation from quinolone or tetracycline antibiotics and levothyroxine.
D3 K2 Lipo Spray (Energetix)	Per label direction	Sublingually in the morning. Hold 30 seconds before swallowing. Increase to the higher label-directed dose if serum 25-OH vitamin D is below 40 ng/mL.
Collagen Complex (Professional Health Products)	Per label direction	Any time of day. Mix in water, juice, or a smoothie.
Lymph-Tone I (Energetix)	Per label direction	Weeks 1 through 4 only. First phase of the drainage sequence.
Lymph-Tone II (Energetix)	Per label direction	Weeks 5 through 8 only. Second phase.
Lymph-Tone III (Energetix)	Per label direction	Weeks 9 through 12 only. Third phase.

Senolytic Synergy

Provides fisetin at 100 mg, quercetin at 200 mg, curcumin C3 Complex at 200 mg, Vinia at 200 mg, and Senactiv at 100 mg per serving.

Fisetin was identified in a Mayo Clinic compound screening as the most potent natural senolytic tested.[10] Quercetin was the plant-derived half of the first published human senolytic trial.

Why it is pulsed rather than taken daily. The proposed senolytic mechanism is hit and run: a short burst at sufficient concentration triggers the elimination of senescent cells, followed by a rest period during which clearance happens. A 2025 review of senolytics and senomorphics states directly that dosing interval, intermittent versus continuous, influences both efficacy and adverse events.[11] Continuous daily dosing is not how these compounds were studied, does not improve results, and may work against the mechanism.

The evidence tier, since this is the product the market shouts loudest about. Laboratory work, animal work, and a small number of early human studies. There is no trial showing that fisetin or quercetin changes how a person on a GLP-1 feels or functions. I use it anyway, pulsed, sequenced behind open drainage, and stopped at ninety days, because the mechanism is coherent and the risk profile at this dose and this duty cycle is low. What I will not do is tell you it is proven.

Orthomega 820 (Orthomolecular)

Provides 430 mg EPA and 390 mg DHA per soft gel, 820 mg combined, as re-esterified triglycerides.

EPA and DHA reduce NF-kB activity and suppress the core SASP cytokines IL-6, TNF-alpha, and IL-1beta. They also address lipid peroxidation, which accelerates directly during rapid fat mobilization. The typical Western omega-6 to omega-3 ratio of fifteen to twenty to one is present in essentially everyone arriving for this program, which is the practical reason this sits in the core rather than in the enhanced layer.

The honest size of the muscle effect. A 2023 meta-analysis found omega-3 supplementation produced a very small but statistically significant benefit to muscle strength (SMD 0.12, 95 percent CI 0.006 to 0.24), with no meaningful effect on muscle mass or on function.[12] The most encouraging single trial ran in healthy adults aged 60 to 85 taking about 3.4 grams of combined EPA and DHA daily for six months, who gained 3.6 percent thigh muscle volume, 2.3 kilograms of grip strength, and 4 percent on one-rep-max strength against control.[13] That trial ran at roughly four times this product's per-serving dose. If muscle preservation is your priority, that gap is worth raising with me rather than assuming one serving a day covers it. Omega-3 is a garnish here. Protein and training are the meal.

Magnesium Bisglycinate (DFH)

Magnesium is a required cofactor for glutathione synthetase, the enzyme that converts cysteine into glutathione. This is why it sits in the core rather than the enhanced layer: **if magnesium is short, NAC 900 in the enhanced layer becomes substantially less effective**, and the entire antioxidant strategy is undermined before it starts.

Bisglycinate is selected over oxide and citrate for absorption. Evening dosing is chosen because the bulk of cellular repair happens during sleep, and magnesium supports sleep quality.

D3 K2 Lipo Spray (Energetix)

Provides 5,000 IU vitamin D3 and 80 mcg K2 (as MK-4 and MK-7) per three-spray serving, delivered sublingually.

Low vitamin D is independently associated with accelerated biological aging, shortened telomere length, elevated SASP markers, and immune senescence. K2 directs calcium appropriately during the rapid body composition changes that characterize this period. Sublingual delivery is chosen to bypass the absorption variability that comes with oral capsules when gastric emptying is slowed, which on a GLP-1 it is by design.

This is the one core product whose dose is directly lab-adjusted. See the marker table below.

Collagen Complex (Professional Health Products)

Contains BioCell Collagen, Type II collagen, chondroitin sulfate, hyaluronic acid, and Opti-MSM to support connective tissue synthesis.

Two things happen at once here. Reactive oxygen species directly cleave collagen cross-links, so degradation speeds up. And the proposed mechanism on the other side is that GLP-1 receptor stimulation on adipose-derived stem cells reduces estrogen production from dermal white adipose tissue, and estrogen is a primary signal driving fibroblast collagen synthesis, so synthesis slows down.[1] Accelerated breakdown and reduced building, in the same tissue, at the same time.

The clinical logic for bioactive collagen peptides is that they signal fibroblasts that breakdown is occurring, which triggers compensatory synthesis. I am labeling that as the reasoning it is rather than dressing it as a trial result, because I could not find the trial. What I can tell you is that this is a protein source you can drink at a moment when getting protein down is the hard part, and that alone earns it a place.

The drainage sequence, and an honest word about the remedies

The lymphatic system is the clearance route for the debris released when senescent cells are eliminated. Lymph has no pump of its own and moves on muscle contraction, which is part of why walking is not filler advice here.

The drainage products in this protocol are homeopathic remedies, and I am not going to be coy about what that means. The research base does not support homeopathy by the standard I have applied to everything else in this guide, and you should hear that from me rather than find it out later. I use these remedies clinically anyway. What I have observed, and I am labeling this as clinical observation rather than trial evidence, is that people who run drainage support alongside the clearance work have an easier first two weeks than people who skip it. The remedies are inexpensive relative to the rest of the protocol and the risk is low. Remedy selection is individualized, which is exactly the kind of thing worth a conversation rather than a purchase decision made from a table.

The sequence is not optional and the remedies are not interchangeable. Each Lymph-Tone runs by itself. Never run two of them at the same time, and never run any one of them continuously for the whole program.

Program weeks	Active	What that phase is doing
Weeks 1 through 4	Lymph-Tone I only	Acute surface-level drainage. Opens clearance ahead of the senolytic debris load.
Weeks 5 through 8	Lymph-Tone II only	Core extracellular matrix clearance. The primary drainage window.
Weeks 9 through 12	Lymph-Tone III only	Deep drainage for longstanding, chronic-level burden.

Each remedy stops when the next one begins. Lymph-Tone I ends at week four, Lymph-Tone II ends at week eight, and Lymph-Tone III carries weeks nine through twelve. Nothing in this sequence overlaps and nothing runs continuously for the full ninety days. Dosing follows each remedy's own label until I set yours with you.

The senolytic pulse calendar

Two capsules on three consecutive days each week, followed by four rest days.

You pick which three days fit your schedule. Once picked, they stay the same for all thirteen weeks. **The consistency of the interval matters more than which days you choose.**

Situation	What to do
Weeks 1 through 13	Two capsules on three consecutive days. Four rest days. Repeat every week without interruption for the full program.
You miss a pulse day	Complete the remaining days of that pulse and resume the normal schedule the following week. **Do not double the dose to compensate.**
You miss an entire week	Resume the normal pulse schedule the following week. Do not attempt to make up the missed pulse.

Layer two: the enhanced protocol

Nothing on this list is added at the start. Each item waits for the result that justifies it, which comes back at the week two call.

Product	Dose	When	What has to be true first
NAC 900 (DFH)	Per label direction	Morning and evening with meals	GGT above 17 IU/L **or** whole blood glutathione below 900 mcmol/L
Liposomal Glutathione (Quicksilver Scientific or Researched Nutritionals)	Per label direction	Morning only, empty stomach. Hold 30 seconds sublingually. Wait 20 minutes before eating.	Same trigger as NAC 900. These two go together.
Nanoemulsified CoQ10 (Quicksilver)	Per label direction	Morning and evening sublingually	Plasma CoQ10 below 0.5 mcg/mL
NAD Gold+ (Quicksilver Scientific)	Per label direction	Morning, with or without food	NAD+ below 20 mcmol/L **or** NAD+/NADH ratio below 3:1
Curcuplex-95 (Xymogen)	Per label direction	Morning, with or without food	Intake questionnaire score 24 to 36, or high symptom burden
Phyto Rad (Energetix)	Per label direction	Morning and evening in warm water	Added alongside Curcuplex-95. Same trigger.
Nanoemulsified DIM (Quicksilver Scientific) (female only)	Per label direction	Sublingually once daily. Hold 30 seconds.	Female, reporting hormonal symptoms, accelerated skin aging, or mood changes
Core Maca Gold (Energetix) (male only)	Per label direction	Morning and evening in warm water	Male, reporting equivalent symptoms. Replaces Nanoemulsified DIM. Never substitute one for the other.
Adaptocrine (Apex Energetics)	Per label direction	Morning, with or without food	HOMA-IR above 2.5

NAC 900 and Liposomal Glutathione

These two are added together and they do different jobs. NAC supplies cysteine, the rate-limiting amino acid in glutathione synthesis, so your body builds its own. Liposomal glutathione supplies the finished antioxidant directly, bypassing the digestive degradation that makes standard oral glutathione capsules largely ineffective. One addresses the substrate shortage, the other addresses the immediate deficit.

Magnesium has to be in place first. NAC leans on the same synthetase step magnesium is required for, which is why magnesium is core and this pair is not.

Screening. NAC may potentiate the anticoagulant effect of warfarin and antiplatelet therapy. This has to be disclosed to your prescriber, who may adjust it or rule it out entirely if you are on active anticoagulation.

Nanoemulsified CoQ10 (Quicksilver)

CoQ10 is the primary electron carrier in the mitochondrial electron transport chain. When it runs low, electrons back up in the chain, which itself generates more reactive oxygen species, which is a loop that feeds itself.

The plausible reason it falls in this population: GLP-1 therapy alters cholesterol synthesis and lipid flux through the mevalonate pathway, which is the same pathway statins block.[3] I am calling that a biologically plausible mechanism rather than a demonstrated one. Liposomal delivery is chosen over capsules on absorption grounds. CoQ10 depletion frequently shows up alongside glutathione depletion, so both get read together.

NAD Gold+ (Quicksilver Scientific), and the argument against it

This one gets a longer explanation than the others, because *Unlock Your GLP-1* deliberately declines to recommend routine NAD+ supplementation and I am not going to pretend that tension does not exist.

The biology is real. SASP induces the enzyme CD38 in neighboring healthy cells, and CD38 degrades NAD+. Senescent cells actively drain the energy currency of the tissue around them, and NAD+ is required for DNA repair, sirtuin activity, and mitochondrial function.[14] NMN is the direct precursor your body uses to make NAD+.

The evidence is thinner than the marketing. The human trials that exist are small and measure laboratory markers rather than anything a person can feel or that predicts health years out.[15]

And there is an open safety question that matters specifically to you. A 2024 analysis linked the terminal breakdown products of excess niacin to major adverse cardiac events across multiple cohorts, with laboratory work showing one of those metabolites directly triggers blood vessel inflammation.[16] That is an association plus a mechanism. It is not proof that an NAD+ precursor harms anyone, and I will not report it to you as though it were. But niacin, NMN, and related compounds clear through a shared pathway, and if you are on a GLP-1 you are almost by definition someone whose cardiovascular risk is under active management.

So here is where I land, and why it is not a contradiction. The ebook declines *routine* NAD+ supplementation, and it is right to. What this protocol does is different: NMN is never added on spec, only when a measured NAD+ result has actually crossed a line, and it is stopped and rechecked at ninety days rather than continued out of habit. If your NAD+ is below 15 mcmol/L, the protocol allows a consideration of a higher label-directed dose with retest at 90 days, and that is a decision I want to make with you in a conversation, not one you should make from a table. If you have known cardiovascular disease or an active vascular concern, this is the product I am most willing to leave out.

Curcuplex-95 (Xymogen) and Phyto Rad (Energetix)

These two are symptom-directed rather than lab-directed, which is a real distinction and worth naming.

Curcumin suppresses NF-kB, the master transcription factor driving SASP cytokine production. That makes it a senomorphic: it reduces the inflammatory output of senescent cells without necessarily eliminating them, which is a different job from the senolytic in the core layer.[11] It uses BCM-95, a bioavailability-enhanced curcumin extract standardized to 65 percent curcumin (roughly 325 mg per capsule from a 500 mg turmeric extract), chosen because standard curcumin bioavailability is poor.

Phyto Rad is a spagyrically processed botanical blend of milk thistle, ginkgo, turmeric, green tea extract, quercetin, and grape seed extract.

These are added together when symptom burden is high enough that broad senomorphic support is worth running independent of any specific confirmed deficiency. I have no trial to hand you for that pairing in this population. What I have is a coherent mechanism, a low risk profile, and a ninety day endpoint at which I read your own log with you and decide whether it earned its place.

Nanoemulsified DIM (Quicksilver Scientific) and Core Maca Gold (Energetix)

These are sex-specific and they are not interchangeable. Do not substitute one for the other.

DIM supports estrogen metabolism in female patients. Adipose tissue is a significant site of peripheral estrogen production, and losing it quickly disrupts that equilibrium. Core Maca Gold is used in male patients to support hypothalamic function and testosterone-pathway health, without the aromatase-related considerations that make DIM inappropriate in men.

Screening. DIM modulates cytochrome P450 enzymes involved in estrogen metabolism, which creates a theoretical interaction with hormonal contraceptives and hormone replacement therapy. Disclose it if either applies to you.

This is clinical reasoning, and I am labeling it as such. If your hormonal symptoms are significant rather than mild, that is a conversation about testing and about whether a prescriber needs to be involved, not a reason to take more of a supplement.

Adaptocrine (Apex Energetics)

Insulin resistance independently drives cellular senescence, which means an elevated HOMA-IR on a GLP-1 is a compounding factor rather than a stray number. Adaptocrine is a multi-herb adaptogen blend built around Panax ginseng and ashwagandha, both with real application in insulin sensitivity. The strongest single piece of clinical evidence is a 2020 systematic review and meta-analysis of ashwagandha in diabetes mellitus, pooling 5 clinical studies, which found the herb restored blood glucose, HbA1c, and insulin toward normal but whose own authors call the clinical base not yet robust enough for strong conclusions.[27] I am naming that tier honestly rather than overselling it.

The interaction that matters here, and it applies to nearly everyone reading this section. Panax ginseng has an established glucose-lowering pharmacology of its own, documented since the 1980s.[28] Layered on top of a medication that is also lowering your glucose, the combined effect can run lower than either one alone. If you are on metformin, a sulfonylurea, insulin, or any other glucose-lowering medication alongside your GLP-1, this goes on the list you review with your prescriber before you start, and your blood glucose gets watched more closely in the first two weeks. See the interaction table below.

The load-bearing part of this response is not the capsule. When HOMA-IR comes back above 2.5, three things happen: your prescriber is notified, dietary carbohydrate reduction and resistance exercise get the emphasis at the week two call, and this product is added. In that order of importance. Retest at 90 days.

The labs that gate layer two

Understanding Your GLP-1 Labs in this kit covers what each of these markers actually measures, what it cannot tell you, and how to get the draw right. Every marker in the table below is covered there. This table is only the decision rule.

A word on how to read them. These are functional markers, which means they are suggestive rather than diagnostic. A result consistent with depletion is a reason to act, not a diagnosis. And where a result lands within ten to fifteen percent of a threshold, your symptom burden decides the call rather than the decimal: a borderline glutathione result in someone with significant fatigue and skin changes gets the enhanced products, a borderline result in someone barely symptomatic waits for the 90 day retest.

Marker	Where it comes from	Threshold	What happens
GGT	LabCorp. On the oxidative panel and on a CMP.	Above 17 IU/L. Optimal range 10 to 17 IU/L.	Add NAC 900 and Liposomal Glutathione. An elevated GGT reflects active glutathione turnover under oxidative burden and is worth acting on even when direct glutathione has not yet fallen out of range.
Glutathione, whole blood	LabCorp. Oxidative panel.	Below 900 mcmol/L	Add NAC 900 and Liposomal Glutathione, regardless of GGT.
CoQ10, plasma	LabCorp. Oxidative panel.	Below 0.5 mcg/mL	Add Nanoemulsified CoQ10, per label direction twice daily sublingually. Read glutathione and GGT alongside it, since these tend to fall together.
NAD+	US BioTek NAD Profile. Dried blood spot, collected at home.	Below 20 mcmol/L, or NAD+/NADH ratio below 3:1	Add NAD Gold+, per label direction daily. Consider a higher label-directed dose if NAD+ is below 15 mcmol/L, with retest at 90 days. Read the NAD Gold+ section above first.
Vitamin D, 25-OH	LabCorp. Ordered with the metabolic panel.	Below 40 ng/mL	Increase D3 K2 Lipo Spray (Energetix) to the higher label-directed dose. Target range 40 to 80 ng/mL. Retest at 90 days. If below 20 ng/mL, that goes to your prescribing physician for a decision about therapeutic dosing, which is not something I direct.
HOMA-IR	LabCorp. Calculated from fasting insulin and glucose on the metabolic panel.	Above 2.5	Add Adaptocrine, per label direction. Notify your prescriber, especially if you take a glucose-lowering medication. Carbohydrate reduction and resistance exercise get the emphasis at the week two call. Retest at 90 days.
Albumin	LabCorp. Included in a CMP.	Below 4.0 g/dL	Protein intake is the urgent item at the week two call. Confirm you are actually meeting the 1.6 g/kg daily minimum. Add collagen to every meal. Recheck albumin at 45 days. If below 3.5 g/dL, that goes to your prescriber for review.
Intake questionnaire	Completed at the start of the program	Score 24 to 36	Add Curcuplex-95 and Phyto Rad regardless of laboratory values.

Marker	Where it comes from	Threshold	What happens
Hormonal symptom burden	Clinical assessment at the intake call	Significant skin aging acceleration, mood changes, or hormonal symptoms	Add Nanoemulsified DIM for female patients or Core Maca Gold for male patients. Symptom-directed, not lab-directed.

Note the HOMA-IR line, because it is the one most often done badly elsewhere. HOMA-IR is calculated from **fasting insulin** alongside fasting glucose. A blood sugar workup that checks glucose and HbA1c without fasting insulin is missing the earliest signal, because insulin rises for years before glucose does.

Reading GGT and glutathione together

These two answer different questions. Direct glutathione reports your current reserve. GGT reports how hard your body is working to keep that reserve filled. Together they say more than either alone.

GGT	Glutathione	What that combination means
Above 17 IU/L	Below 900 mcmol/L	Maximum oxidative burden. Defenses depleted and actively under stress. Full NAC and GSH support, and this is the highest priority version of the enhanced layer.
Above 17 IU/L	At or above 900 mcmol/L	Compensated oxidative stress. You are keeping pace, but working hard to do it. NAC and GSH go in to prevent depletion before it happens.
17 IU/L or below	Below 900 mcmol/L	Depletion without current acute oxidative activity. NAC and GSH at standard dosing.
17 IU/L or below	At or above 900 mcmol/L	Antioxidant status adequate. Hold NAC and GSH unless symptom burden is high. Core magnesium keeps supporting your own glutathione synthesis. Reassess at the 90 day retest.

Thiamine, the short course that runs on its own clock

Thiamine gets its own section because *Unlock Your GLP-1* treats it as one of the most important nutrients in the whole picture, and because it runs as a short repletion course on its own clock rather than living inside either layer.

Why it behaves differently from everything else here. In the trial that worked out how semaglutide actually produces weight loss, total energy intake across a day fell by 24 percent.[17] Your vitamin and mineral requirement did not fall by 24 percent, and reviews now describe a recognizable pattern of micronutrient shortfalls following sustained low intake on GLP-1 therapy.[18] Thiamine is the sharpest example, because your body holds only a small reserve of it. Status falls within weeks of poor intake, not months. In one obesity medicine clinic population, about a third of patients already tested thiamine deficient before any surgery or medication.[19] Low thiamine is also common in type 2 diabetes.[20]

Bariatric surgery produces the same combination of sharply reduced intake plus vomiting, and precisely because of that it has an established thiamine monitoring protocol.[21,22] Reviews are now asking directly whether GLP-1 care should borrow that framework rather than reinvent it.[21]

Keep the emergency in proportion, and keep it in mind. A systematic review found six published cases of Wernicke's encephalopathy following semaglutide for obesity,[23] and an analysis of FDA adverse event data identified fifteen cases across GLP-1 medications as a whole.[24] Against many millions of prescriptions, that is a very small number. Two details from those reports are worth carrying with you anyway. Nearly every case had prolonged gastrointestinal symptoms or substantial rapid weight loss first, which means there was a warning period. And among patients followed afterward, lasting neurological problems were common, which is what makes recognizing it early matter so much.

Which form, and the honest read on it. Benfotiamine raises thiamine levels more effectively than standard thiamine hydrochloride.[25] Better absorption is not the same as better results. The longest trial to date gave benfotiamine for twelve months to people with type 2 diabetes and symptomatic nerve pain: it raised every thiamine marker measured and was well tolerated, and it produced no significant improvement in nerve structure, function, or symptoms against placebo.[26] Read honestly, that makes benfotiamine a proven way to raise thiamine status and a poor bet as a treatment for established neuropathy. Raising status is exactly the job it has here, and benfotiamine is the form I use for it.

The dose I use. Benfotiamine, 150 mg twice daily, for 4 to 6 weeks. This is repletion rather than maintenance, which is why it carries a stop point instead of running for the full program. Thiamine is also measurable, and testing beats guessing: *Understanding Your GLP-1 Labs* in this kit covers the draw, and if your intake has been steady and your status comes back fine, this is a course you get to skip. One more honest sentence, because the framing in this guide does not change for thiamine: no outcome trial has tested this course in people on a GLP-1 medication. I use it because the shortfall it corrects is real, measurable, and inexpensive to fix.

Your day, hour by hour

Time	What goes in
Morning, empty stomach	Liposomal Glutathione if you are on it. Wait 20 minutes before eating or taking anything else.
Morning, with or without food	D3 K2 Lipo Spray (Energetix) sublingually. The active Lymph-Tone remedy, per its label direction. Then, if prescribed: Nanoemulsified CoQ10 sublingual, NAD Gold+, Curcuplex-95, Phyto Rad in warm water, Adaptocrine, Nanoemulsified DIM sublingual for female patients.
Morning, with a meal	NAC 900 if prescribed.
Largest meal of the day	Orthomega 820.
Any time	Collagen Complex in water, juice, or a smoothie.
Evening, with a meal	NAC 900 second dose if prescribed.
Evening	Magnesium Bisglycinate (DFH). The active Lymph-Tone remedy again, if its label directs a second daily dose. Then, if prescribed: Nanoemulsified CoQ10 second dose, Phyto Rad second dose, Core Maca Gold second dose for male patients.
Morning and evening, only during the thiamine course	Benfotiamine, 150 mg each time, for the 4 to 6 week course described above.
Pulse days only	Senolytic Synergy, two capsules. Three consecutive days per week.

Handling and storage. Shake all tinctures and liposomal products before each use. Store away from direct sunlight and heat, which means not in a vehicle and not near a stove. Liposomal Glutathione goes on an empty stomach with a 20 minute wait before eating. Senolytic Synergy should not be refrigerated, store it at room temperature. Hold all sublingual sprays and tinctures under the tongue for 30 seconds before swallowing. **Do not combine tinctures in the same glass.** Give each one its own small volume of warm water.

The first two weeks, and how to read them

Some people experience transient fatigue, mild joint achiness, or subtle cognitive fog during weeks one and two. The explanation usually offered is that senolytic activity is releasing the stored inflammatory contents of the cells being eliminated, and that the acute drainage support running concurrently is there specifically to move that debris out. It typically settles within several days.

One caution about that framing, because it matters more than the framing does. If feeling better proves the protocol is working, and feeling worse also proves the protocol is working, then the claim cannot be tested, and an untestable claim is a poor reason to keep going. So I treat worsening as information, never as a milestone. Sometimes it is the clearance response and it passes. Sometimes it means the dose is too high, or the drainage and hydration groundwork was skipped, or this is the wrong approach for you.

Symptoms that are severe, that persist beyond two weeks, or that first appear at week four or later are a different event and get handled differently. Early clearance reactions and late-onset symptoms are not the same thing, and late-onset symptoms are a reason to reassess with me rather than to push through.

Take a baseline before you change anything. Energy out of ten, sleep out of ten, and how you feel in the two days after a training session. Then judge this against your own log rather than against anyone's promise, including mine.

Interactions and exclusions to clear before you start

Your complete medication and supplement list gets reviewed before anything begins. Bring all of it, including anything that comes in a blend.

Product	The interaction	What happens
NAC 900	May potentiate the anticoagulant effect of warfarin and antiplatelet therapy.	Disclose to your prescriber. Dose adjusted, or excluded outright if you are on active anticoagulation.
Orthomega 820	Additive antiplatelet effect with direct oral anticoagulants at therapeutic doses.	Disclose to your prescriber. Dose reduced, or excluded if you are on active anticoagulation.
Magnesium Bisglycinate	Reduces absorption of quinolone and tetracycline antibiotics and levothyroxine.	Minimum two hour separation, every time. This one is easy to get wrong and it matters.
D3 K2 Lipo Spray	Thiazide diuretics reduce renal calcium excretion, and high dose vitamin D alongside a thiazide raises the risk of hypercalcemia.	Vitamin D status confirmed by laboratory before any dose increase. Dose capped if you are on a thiazide.
Nanoemulsified DIM	Modulates cytochrome P450 enzymes involved in estrogen metabolism. Theoretical interaction with hormonal contraceptives and HRT.	Disclose if applicable. Female patients only. Never substituted into a male protocol.
Adaptocrine	Panax ginseng has a documented hypoglycemia interaction with antidiabetic medication (case report: hypoglycemia in a patient on metformin and sitagliptin).	Disclose any glucose-lowering medication to your prescriber before starting. Blood glucose monitored more closely in the first two weeks.

Absolute exclusions. Pregnancy. Nursing. If you become pregnant during the program, stop the enhanced layer immediately and contact both me and your obstetric provider.

And the line that governs all of it. This protocol does not replace GLP-1 prescribing management. You continue your prescribed dosing without modification. Anything about the medication itself belongs with the physician who prescribes it.

Where to get these, and what I earn

Look for third-party testing first. NSF Certified for Sport and USP Verified mean an independent laboratory confirmed what is in the bottle. That is not automatic in this industry.

Several products named above come from practitioner lines rather than retail shelves, which is a fact about how they are distributed and not a claim that they are better than what you can buy elsewhere. Some of those lines are not carried in my dispensary, so do not assume the link below covers the whole protocol. Where a retail equivalent exists and is third-party tested, it is a reasonable substitute, and the doses in this guide are the thing to match. Where you cannot find one, bring me the product name at the discovery session and I will tell you what I would use in its place rather than leave you hunting for it.

For screened versions of the products that are carried there, my dispensary is here:

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases made through that link, and Root Health has a financial interest in the products used in this program. Nothing in this guide was included, excluded, ranked, or dosed based on what it earns. The two interventions I have put ahead of every product in this document, protein and resistance training, earn nothing at all. You can buy any of this anywhere.

What I deliberately left out, and why

Any instruction about your GLP-1 medication. Not the dose, not the escalation schedule, not whether to pause it. That is your prescriber's decision and it stays there. If something in this guide makes you want to change your prescription, bring this guide to that appointment instead.

Anything sold as reversing biological age. Biological age tests are estimates built from models, the models disagree with each other, and the useful version of that test is running the same one twice and comparing it to itself.

High-dose proprietary blends with undisclosed amounts, and any product whose interpretive framework exists only in the marketing material of the company selling it.

A guide sold to someone I have never examined has no history, no medication list, and no way to change course next week. That is why the bar for what goes into it is higher than the bar for what I would discuss with someone in front of me, and it is the reason you can trust the rest of this document.

When to seek clinical review

Urgently, the same day: confusion, unsteadiness, or vision changes after a stretch of persistent vomiting or very low intake. Go to an emergency department and say “possible thiamine deficiency.”

Promptly, with your prescriber: persistent vomiting or an inability to keep food down, rapid weight loss that is accelerating rather than steadying, or new severe abdominal pain.

Before you start anything here: if you take an anticoagulant or antiplatelet medication, levothyroxine, a thiazide diuretic, hormonal contraception, or hormone replacement therapy.

At your next appointment: if you are on a protocol with no defined endpoint, or if the symptoms in Key 3 of the ebook are severe or worsening rather than mild and stable.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=glp1-supplement

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

You can also reach the office at (919) 662-0520.

Bring your medication list, your most recent labs, and how long you have been on your medication. The two-layer structure in this guide only works if someone actually reads your numbers against it, and that conversation is free.

Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=glp1-supplement

If you would rather start on your own, take the baseline first: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=glp1-supplement

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